Master's Thesis

Deep Reinforcement Learning with State Restoration Model for Bridging the Sim-to-Real Gap in Two-Wheeled Bipedal Robot

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硕士论文

基于状态恢复模型的深度强化学习在双轮双足机器人仿真到现实差距弥合中的应用

Jae hyung Cho (조 재 형) 融合IT工程系
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상태 복원 모델 기반 심층 강화학습 제어기를 이용한
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A thesis submitted to the faculty of the Pohang University of Science and Technology in partial fulfillment of the requirements for the degree of Master of Science in the IT Engineering

Pohang, Korea
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The undersigned have examined this thesis and hereby certify that it is worthy of acceptance for a master's degree from POSTECH

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Deep Reinforcement Learning with State Restoration Model for Bridging the Sim-to-Real Gap in Two-Wheeled Bipedal Robot

基于状态恢复模型的深度强化学习在缩小两轮平衡机器人仿真与现实差距中的应用

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ABSTRACT

Deploying locomotion policies on real robots remains challenging because onboard sensing provides only partial observations and because policies trained in simulation often degrade on hardware due to the sim-to-real gap. This thesis investigates sim-to-real deep reinforcement learning for a two-wheeled bipedal robot performing hybrid wheeled–legged locomotion, where coupled wheel–leg dynamics and intermittent contacts induce partial observability, making instantaneous measurements insufficient to form a Markovian state for stable control.

We propose a State Restoration Model (SRM) that reconstructs an enriched, state-like latent representation from observation histories. The SRM is trained to approximate privileged full-state information available in simulation and is integrated into an asymmetric actor–critic training pipeline: the actor is conditioned on the restored representation, while the critic leverages privileged information during training.

We evaluate the proposed approach on FlaminGO, an 8-DoF two-wheeled bipedal robot, following a simulation → simulation → real pipeline: training in Isaac Sim with Isaac Lab, cross-simulator validation in MuJoCo, and deployment on hardware. Across cross-simulator and real-world experiments, SRM improves commanded-velocity tracking and reduces steady-state bias compared with a baseline trained without SRM, including more stable station-keeping behavior under near-zero commands. These results

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摘要

在真实机器人上部署运动策略仍然具有挑战性，因为机载传感器仅能提供部分观测，并且由于仿真到现实差距，在仿真中训练的策略在硬件上往往会性能下降。本论文研究针对执行混合轮式-腿式运动的双轮双足机器人的仿真到现实深度强化学习，其中耦合的轮-腿动力学和间歇性接触导致了部分可观测性，使得瞬时测量不足以形成用于稳定控制的马尔可夫状态。

我们提出了一种状态恢复模型 (SRM)，该模型从观测历史中重建一个增强的、类似状态的潜在表示。SRM 经过训练以近似仿真中可用的特权全状态信息，并集成到非对称演员-评论家训练流程中：演员网络以恢复的表示为条件，而评论家网络在训练期间利用特权信息。

我们在FlaminGO（一款8自由度两轮双足机器人）上评估了所提出的方法，遵循仿真 → 仿真 →真实的流程：在Isaac Lab的Isaac Sim中进行训练，在MuJoCo中进行跨模拟器验证，并在硬件上部署。在跨模拟器和真实世界实验中，与未使用SRM训练的基线相比，SRM改善了指令速度跟踪并减少了稳态偏差，包括在接近零指令下更稳定的定点保持行为。这些结果

suggest that restoring a state-like representation from temporal context can mitigate ambiguity induced by partial observability and improve sim-to-real transfer robustness for hybrid locomotion.

表明，从时间上下文中恢复类状态表示可以缓解部分可观测性引起的模糊性，并提高混合运动的仿真到现实迁移鲁棒性。

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I. Introduction

Deploying locomotion policies on real robots is challenging because onboard observations often do not fully represent the underlying state required for stable control. This *partial observability* is commonly formalized as a partially observable Markov decision process (POMDP), where robust decision-making generally requires integrating information over time rather than relying on a single instantaneous observation [1]. In addition, policies trained in simulation frequently degrade on hardware due to the *sim-to-real gap*, which arises from discrepancies such as unmodeled dynamics, sensor noise, delays, actuator bandwidth limits, and parameter variations [2, 3]. Together, partial observability and sim-to-real mismatch pose fundamental obstacles to reliable real-world locomotion [4, 5].

These obstacles are particularly salient for hybrid locomotion systems that combine wheels and legs. While wheels provide efficient traversal on flat terrain and legs can negotiate discontinuities such as steps and rough ground, combining rolling dynamics with intermittent contacts yields coupled behaviors and transition scenarios that are sensitive to sensing and terrain variations [6]. As a result, many hybrid systems rely on mode-dependent heuristics (e.g., separate rolling and stepping controllers) that require extensive tuning and can degrade performance at transitions [6]. Model-based approaches such as Model Predictive Control (MPC) [7, 8] further depend on accurate models of contact, friction, actuator dynamics, and latency, and they can be sensitive to modeling errors and unanticipated environment variations.

Deep reinforcement learning (DRL) has emerged as an alternative for learning locomotion behaviors with reduced manual crafting, demonstrating simulation-to-hardware transfer in legged systems [4, 5]. In typical pipelines, a policy is optimized in simulation (often with PPO [9]) and then deployed on hardware. However, robust deployment still hinges on addressing both sim-to-real mismatch and partial observability [2, 3, 1].

To mitigate sim-to-real mismatch, prior work introduces variability during training via domain randomization [2] and dynamics randomization [3]. Partial observability, meanwhile, motivates leveraging temporal context, and recurrent or memory-based RL methods have been used to summarize observation histories in partially observable environments [10]. Building on these ideas, we develop a DRL-based sim-to-real framework for hybrid locomotion that combines domain/dynamics randomization [2, 3] with a **State Restoration Model (SRM)**. SRM reconstructs an enriched, state-like latent representation from observation histories by integrating current sensory inputs with temporal context, reducing ambiguity when observations alone are insufficient to infer a Markovian state [1, 10].

We evaluate the proposed framework on **FlaminGO**, a multi-modal locomotion robot inspired by

I. 引言

在真实机器人上部署运动策略具有挑战性，因为机载观测通常无法完全代表稳定控制所需的基础状态。这种部分可观测性通常被形式化为部分可观测马尔可夫决策过程（POMDP），其中稳健的决策制定通常需要随时间整合信息，而非依赖单一的瞬时观测 [1]。此外，在仿真中训练的策略在硬件上常常性能下降，这是由于仿真到现实差距造成的，该差距源于未建模动力学、传感器噪声、延迟、执行器带宽限制以及参数变化等差异 [2, 3]。部分可观测性与仿真到现实不匹配共同构成了可靠现实世界运动 [4, 5]的根本障碍。

这些障碍对于结合轮子和腿的混合运动系统尤为突出。虽然轮子在平坦地形上提供高效移动，腿可以应对台阶和崎岖地面等不连续地形，但将滚动动力学与间歇接触相结合会产生耦合行为和过渡场景，这些对感知和地形变化非常敏感 [6]。因此，许多混合系统依赖模式依赖启发式方法（例如，分离的滚动和步进控制器），这些方法需要大量调参，并且在过渡时可能性能下降 [6]。基于模型的方法，如模型预测控制（MPC） [7, 8]，进一步依赖于接触、摩擦、执行器动力学和延迟的精确模型，并且可能对建模错误和未预料的环境变化敏感。

深度强化学习（DRL）已成为一种替代方案，用于学习运动行为并减少人工设计，展示了在腿关节系统中的仿真到硬件迁移能力 [4, 5]。在典型流程中，策略在仿真中（通常使用 PPO [9]）进行优化，然后部署到硬件上。然而，稳健的部署仍然依赖于解决仿真到现实不匹配和部分可观测性 [2, 3, 1] 这两个问题。

为了缓解仿真到现实不匹配，先前的工作通过在训练中引入域随机化 [2] 和动力学随机化 [3] 来增加变异性。同时，部分可观测性促使利用时间上下文，循环或基于记忆的强化学习方法已被用于在部分可观测环境中总结观测历史 [10]。基于这些思路，我们开发了一个基于深度强化学习的仿真到现实框架，用于混合运动，该框架结合了域/动力学随机化 [2, 3] 与一个**状态恢复模型 (SRM)**。SRM 通过整合当前感官输入与时间上下文，从观测历史中重建一个增强的、类状态潜在表示，从而在仅凭观测不足以推断马尔可夫状态时减少歧义 [1, 10]。

我们在受火烈鸟关节结构启发的多模态运动机器人**FlaminGO**上评估了所提出的框架（图1.1）。

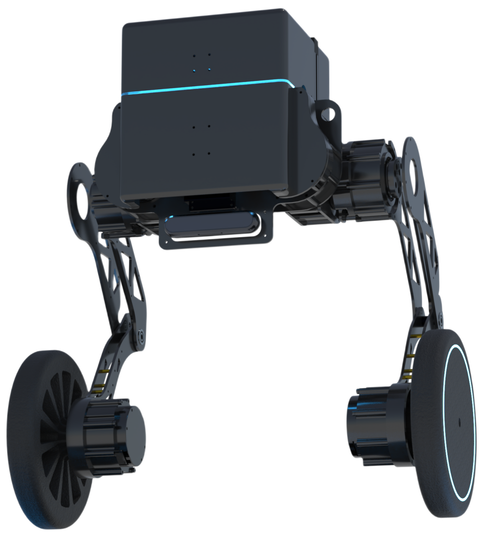

Figure 1.1: The FlaminGO robot: A novel multi-modal locomotion robot with hybrid wheeled and legged mechanisms. Each side of the robot consists of 4 degrees of freedom (DoF)—hip, shoulder, leg, and wheel joints—resulting in a total of 8 DoF.

the joint structure of flamingos (Fig. 1.1). FlaminGO integrates wheeled mobility with leg articulation for negotiating stairs and uneven ground. Each side consists of 4 degrees of freedom resulting in a total of 8 DoF. The actuation system uses only 8 actuators, compared to the 12 typically found in quadrupedal robots [11, 12, 13].

The contributions of this work are summarized as follows:

- **Multi-modal robot design:** We design and implement FlaminGO, an 8-DoF two-wheeled bipedal robot that combines wheeled mobility and leg articulation for hybrid locomotion across diverse terrains.
- **DRL-based sim-to-real framework:** We establish a simulation training and transfer pipeline for hybrid locomotion and incorporate domain/dynamics randomization to reduce sim-to-real sensitivity [2, 3].
- **State restoration under partial observability:** We propose SRM to reconstruct a state-like latent representation from observation histories, improving robustness when observations alone do not fully represent the underlying state [1, 10].

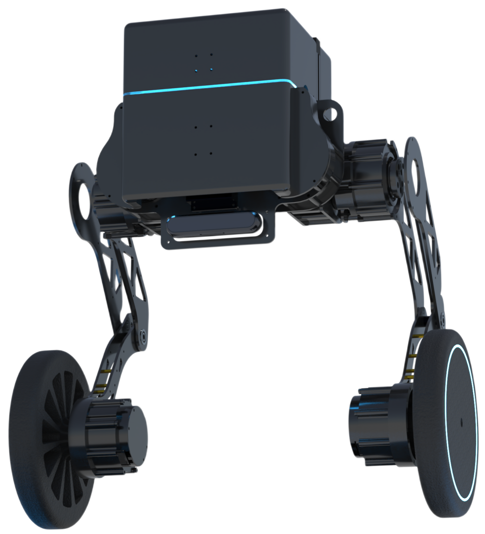

图1.1：FlaminGO机器人：一种采用轮腿混合机构的新型多模态运动机器人。机器人每侧包含4个自由度（DoF）——髋关节、肩关节、腿关节和轮关节——共计8个自由度。

FlaminGO将轮式移动与腿部关节运动相结合，以应对楼梯和不平坦地面。每侧具有4个自由度，总计8个自由度。该驱动系统仅使用8个执行器，而典型的四足机器人通常需要12个 [11, 12, 13]。

本工作的贡献总结如下：

- **多模态机器人设计：**我们设计并实现了FlaminGO，一款8自由度双轮双足机器人，它结合了轮式移动与腿部关节运动，能够在多种地形上实现混合运动。
- **基于DRL的仿真到现实框架：**我们为混合运动建立了一套仿真训练与迁移流程，并引入域/动力学随机化以降低仿真到现实的敏感性 [2, 3]。
- **部分可观测性下的状态恢复：**我们提出SRM方法，从观测历史中重建类状态潜在表示，从而在观测数据无法完全反映底层状态时提升鲁棒性 [1, 10]。

II. Preliminaries

2.1 Reinforcement Learning

To enable hybrid locomotion for the FlaminGO robot, we model the control problem as a Markov Decision Process (MDP) defined by the tuple (S, A, P, R, γ). Here, S represents the robot's kinematic and dynamic states, A denotes the action space comprising motor commands, $P(s'|s,a)$ describes the transition dynamics, $R(s,a)$ is the reward function that evaluates the desirability of a given state-action pair, and γ is the discount factor balancing immediate and future rewards. At each time step t, the robot observes its state s_t, selects an action a_t following policy $\pi(a|s)$, transitions to a new state s_{t+1}, and receives a reward r_t. The objective is to optimize the policy π^* to maximize the expected cumulative reward.

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \tag{2.1}$$

where γ is the discount factor that balances immediate and future rewards.

To address the control problem formulated as an MDP, we employ Proximal Policy Optimization (PPO)[9], a policy-gradient algorithm known for its balance of stability and efficiency in learning. PPO iteratively refines the policy by constraining updates within a safe range, ensuring steady progress during training. Its clipping mechanism mitigates the risk of unstable updates, making it particularly suitable for complex robotic tasks. The PPO objective function is given by:

$$\mathcal{L}^{PPO}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right], \tag{2.2}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ represents the probability ratio, $\hat{A}_t$ is the advantage estimate, and ϵ is the clipping parameter.

2.2 Two-Wheeled Bipedal Robot Design

FlaminGO is a two-wheeled bipedal robot designed for hybrid locomotion by combining wheeled mobility with leg articulation. Each side of the robot has four actuated joints (hip, shoulder, leg, and wheel), resulting in a total of 8 DoF. The hip and shoulder joints are actuated by Robstride RS03[14] units with a gear reduction ratio of 9:1, providing a rated torque of 20.0 Nm and a peak torque of 60.0 Nm. The leg joint uses a Robstride RS03 with an external gear module (gear reduction ratio 13.5:1), with a rated torque of 30.0 Nm and a peak torque of 90.0 Nm. For wheeled propulsion, the wheel joint is driven

II. 预备知识

2.1 强化学习

为了实现FlaminGO机器人的混合运动，我们将控制问题建模为马尔可夫决策过程（MDP），由元组(S, A, P, R, γ)定义。其中，S表示机器人的运动学和动力学状态，A表示包含电机指令的动作空间，$P(s'|s,a)$描述状态转移动力学，$R(s,a)$是评估给定状态-动作对优劣的奖励函数，γ是平衡即时奖励与未来奖励的折扣因子。在每个时间步t，机器人观测其状态s_t，根据策略$\pi(a|s)$选择动作a_t，转移至新状态s_{t+1}，并获得奖励r_t。目标是优化策略π^*以最大化期望累积奖励。

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \tag{2.1}$$

其中γ是平衡即时奖励与未来奖励的折扣因子。

为了解决以MDP形式表述的控制问题，我们采用近端策略优化（PPO）[9], 一种策略梯度算法，以其在学习过程中兼顾稳定性与效率而著称。PPO通过将更新约束在安全范围内来迭代优化策略，确保训练过程中的稳定进展。其裁剪机制降低了不稳定更新的风险，使其特别适用于复杂的机器人任务。PPO的目标函数如下:

$$\mathcal{L}^{PPO}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right], \tag{2.2}$$

其中 $r_t(\theta) = \frac{\pi_\theta}{(a_t|s_t)}\pi_{\theta_{\text{old}}(a_t|s_t)}$ 表示概率比，$\hat{}$ A_t是优势估计值，ϵ是裁剪参数。

2.2 双轮双足机器人设计

FlaminGO 是一款双轮双足机器人，旨在通过结合轮式移动与腿部关节运动实现混合运动。机器人每侧配备四个驱动关节（髋关节、肩关节、腿关节和轮关节），共计8个自由度。髋关节和肩关节由 Robstride RS03[14] 单元驱动，减速比为 9:1，提供额定扭矩 20.0 Nm 和峰值扭矩 60.0 Nm。腿关节采用 Robstride RS03 并配备外部齿轮模块（减速比 13.5:1），额定扭矩为 30.0 Nm，峰值扭矩为 90.0 Nm。在轮式驱动方面，轮关节由

by a Robstride RS06[14] actuator with a 9:1 gear reduction ratio, delivering a rated torque of 12 Nm and a peak torque of 36 Nm. The onboard computation is performed on an NVIDIA Jetson Orin Nano 8GB single-board computer (SBC), and inertial sensing is provided by an AHRS EBIMU-9DOFV5-R3 IMU. The robot is powered by a Samsung SDI 21700 50S battery pack configured as 13S1P.

The mechanical design targets hybrid locomotion scenarios in which the robot can roll efficiently on flat terrain and step over discontinuities such as stairs or uneven ground by coordinating leg articulation and wheel actuation. Table 2.1 summarizes the key components and specifications. The base weight of the platform is 10.5 kg, and the total weight including all components is 18.5 kg.

Table 2.1: Component Specifications of the Robot

Component Name	**Specification**
Joint Actuators (Hip, Shoulder)	Robstride RS03 Rated Torque: 20.0 Nm Peak Torque: 60.0 Nm Gear Reduction Ratio: 9:1
Joint Actuators (Leg)	Robstride RS03 (W/ External Gear Module) Rated Torque: 30.0 Nm Peak Torque: 90.0 Nm Gear Reduction Ratio: 13.5:1
Joint Actuators (Wheel)	Robstride RS06 Rated Torque: 12 Nm Peak Torque: 36 Nm Gear Reduction Ratio: 9:1
Single Board Computer (SBC)	NVIDIA Jetson Orin Nano 8GB
Inertia Measurement Unit (IMU)	AHRS EBIMU-9DOFV5-R3
Battery	Samsung SDI 21700 50S 13s 1p
Base Weight	10.5 kg
Total Weight	18.5 kg

Robstride RS06[14] 执行器驱动，减速比为 9:1，提供额定扭矩 12 Nm 和峰值扭矩 36 Nm。机载计算由 NVIDIA Jetson Orin Nano 8GB 单板计算机（SBC）完成，惯性传感由 AHRS EBIMU-9DOFV5-R3 IMU 提供。机器人由配置为 13S1P 的三星SDI 21700 50S电池组供电。

机械设计针对混合运动场景，使机器人能够在平坦地形上高效滚动，并通过协调腿部关节运动与轮式驱动跨越楼梯或不平地面等障碍。表2.1总结了关键组件与规格参数。平台基础重量为10.5千克，包含所有组件后的总重量为18.5千克。

表 2.1：机器人组件规格

组件名称	**规格**
关节执行器（髋关节、肩关节）	Robstride RS03 额定扭矩：20.0 Nm 峰值扭矩：60.0 Nm 减速比：9:1
关节执行器（腿关节）	Robstride RS03（带外部齿轮模块） 额定扭矩：30.0 Nm 峰值扭矩：90.0 Nm 减速比：13.5:1
关节执行器（车轮）	Robstride RS06 额定扭矩：12 Nm 峰值扭矩：36 Nm 减速比：9:1
单板计算机 (SBC)	NVIDIA Jetson Orin Nano 8GB
惯性测量单元 (IMU)	AHRS EBIMU-9DOFV5-R3
电池	三星SDI 21700 50S 13s 1p
底座重量	10.5 千克
总重量	18.5 千克

2.3 Sim-to-Real Transfer

Sim-to-real transfer aims to deploy policies trained in simulation to real-world robots by mitigating the *sim-to-real gap*, i.e., performance degradation caused by discrepancies between simulated and real dynamics (e.g., unmodeled effects, sensing noise, latency, and parameter variations). To reduce this gap, we employ the following techniques:

- **Domain Randomization**: We randomize physical parameters in simulation (e.g., mass properties, friction, actuator gains, and sensor noise) to train policies that generalize across plausible real-world conditions [2].
- **State Restoration Model (SRM)**: To address partial observability, we reconstruct an enriched, state-like latent representation from observation histories using SRM. By integrating current sensory inputs with temporal context, SRM provides the policy with a compact surrogate state that is more informative than instantaneous observations alone, improving robustness under noisy and incomplete measurements.
- **Latent State Transformation**: We transform raw observations into a latent vector using a recurrent belief encoder that aggregates proprioceptive and exteroceptive signals over time. This latent representation supports state estimation under partial observability and enables the policy to incorporate noisy exteroceptive inputs in a structured manner [15].

By combining domain randomization with history-based latent state inference, the resulting policy is trained to tolerate dynamics mismatch and sensing limitations, supporting reliable deployment for hybrid locomotion. The experimental results in later sections evaluate how these components reduce the sim-to-real gap in real-world operation.

2.3 仿真到现实迁移

仿真到现实迁移旨在通过缩小仿真到现实差距，将在仿真中训练的策略部署到真实机器人上。该差距指因仿真与真实动力学之间的差异（例如未建模效应、感知噪声、延迟及参数变化）导致的性能下降。为减小这一差距，我们采用以下技术：

- **域随机化**：我们对仿真中的物理参数（例如质量属性、摩擦力、执行器增益和传感器噪声）进行随机化，以训练能够泛化到各种真实世界条件下的策略 [2]。
- **状态恢复模型 (SRM)**：为了解决部分可观测性问题，我们利用SRM从观测历史中重建一个增强的、类状态潜在表示。通过将当前传感输入与时间上下文相结合，SRM为策略提供了一个紧凑的替代状态，该状态比瞬时观测信息更丰富，从而提高了在噪声和不完整测量下的鲁棒性。
- **潜在状态变换**：我们使用一个循环信念编码器将原始观测转换为潜在向量，该编码器随时间聚合本体感觉和外感受信号。这种潜在表示支持在部分可观测性下的状态估计，并使策略能够以结构化的方式整合有噪声的外感受输入 [15]。

通过结合域随机化与基于历史信息的潜在状态推断，训练出的策略能够容忍动力学不匹配和感知限制，从而支持混合运动的可靠部署。后续章节的实验结果将评估这些组件如何在实际操作中缩小仿真到现实差距。

III. Method

3.1 State Restoration Model

In real-world scenarios, robots often operate in environments where only partial observations of the full state are available due to sensor limitations or unobservable dynamics. This partial observability can lead to suboptimal decision-making and hinder the performance of reinforcement learning (RL)-based control policies, especially when transferring them from simulation to reality. In simulation, privileged information—such as precise physical states or dynamics—can be readily accessed, but these are typically unavailable in real-world deployments. This disparity introduces a significant challenge known as the sim-to-real gap, where policies trained in simulation fail to generalize effectively to the real world.

To address this challenge, we employ an *Asymmetric Actor-Critic* framework[16], where the critic network benefits from privileged information during training to evaluate actions, while the actor network operates solely on observable states. Despite this improvement, the absence of privileged information during deployment can limit the actor's performance.

As illustrated in Figure 3.1, the State Restoration Model (SRM) bridges this gap by reconstructing an enriched state, $\hat{s}_t^f$, from the observable state, s_t^o. This enriched state approximates the privileged full state, s_t^f, ensuring robust and consistent decision-making for the actor network in real-world environments. The SRM is trained using a combination of reconstruction and reward consistency losses, described in subsequent sections.

3.1.1 Architecture and Training Process

The GRU encoder processes the partially observable state to estimate the privileged observations and reconstruct the full state as follows:

$$s_t^p = \mathrm{GRU}_{\mathrm{encoder}}(s_t^o) \tag{3.1}$$

$$\hat{s}_t^f = (s_t^o)||(s_t^p), \tag{3.2}$$

where s_t^o represents the partially observable input at time t, and $\hat{s}_t^f$ is the reconstructed full state. The GRU encoder processes the sequential input s_t^o to capture temporal dependencies and generate the predicted hidden state s_t^p. The reconstructed full state $\hat{s}_t^f$ is then obtained by concatenating s_t^o and s_t^p, serving as an approximation of s_t^f.

III. 方法

3.1 状态恢复模型

在真实世界场景中，由于传感器限制或不可观测的动力学因素，机器人通常只能在获取部分状态观测的环境下运行。这种部分可观测性可能导致次优决策，并阻碍基于强化学习的控制策略的性能，尤其是在从仿真向现实迁移时。在仿真中，特权状态信息（如精确的物理状态或动力学参数）可以轻易获取，但这些信息在真实世界部署中通常不可用。这种差异带来了一个重大挑战，即所谓的仿真到现实差距，导致在仿真中训练的策略难以有效泛化到现实世界。

为应对这一挑战，我们采用了一种非对称演员-评论家框架[16], ，其中评论家网络在训练期间利用特权信息来评估动作，而演员网络仅基于可观测状态运行。尽管有此改进，但在部署期间缺乏特权信息仍可能限制演员网络的性能。

如图3.1所示，状态恢复模型 (SRM) 通过从可观测状态 s_t^o重建一个增强状态$\hat{s}_t^f$ 来弥合这一差距。该增强状态近似于特权完整状态 s_t^f ，确保演员网络在现实环境中做出稳健且一致的决策。SRM 使用重建损失和奖励一致性损失的组合进行训练，具体内容将在后续章节中描述。

3.1.1 架构与训练过程

GRU编码器处理部分可观测状态以估计特权观测，并按如下方式重建完整状态：

$$s_t^p = \mathrm{GRU}_{\mathrm{encoder}}(s_t^o) \tag{3.1}$$

$$\hat{s}_t^f = (s_t^o)||(s_t^p), \tag{3.2}$$

其中 s_t^o 表示时间 t的部分可观测输入， $\hat{s}_t^f$ 是重构完整状态。GRU编码器处理序列输入 s_t^o 以捕获时间依赖性并生成预测隐藏状态 s_t^p。重构完整状态$\hat{s}_t^f$随后通过拼接 s_t^o和 s_t^p 获得，作为 s_t^f 的近似。

The SRM is trained using a restoration-based loss function, which minimizes the discrepancy between the restored state $\hat{s}_t^f$ and the full state s_t^f. The restoration loss is defined as:

$$\mathcal{L}_{\text{reconstruction}}(\hat{s}_t^f, s_t^f) = \begin{cases} \frac{1}{2}(\hat{s}_t^f - s_t^f)^2 & \text{if } |\hat{s}_t^f - s_t^f| \leq 1.0, \\ 1.0(|\hat{s}_t^f - s_t^f| - 0.5) & \text{otherwise} \end{cases} \tag{3.3}$$

ensuring that the reconstructed state closely approximates the privileged information used by the critic network during training.

To further enhance task-specific performance, SRM incorporates a reward consistency loss. This loss minimizes the difference in rewards computed using the full state s_t^f and the reconstructed state $\hat{s}_t^f$, aligning the task-relevant features between the two states:

$$\mathcal{L}_{\text{reward}} = \|R(s_t^f, a_t) - R(\hat{s}_t^f, a_t)\|^2 \tag{3.4}$$

The total loss for training the SRM is expressed as:

$$\mathcal{L}_{\text{SRM}} = \mathcal{L}_{\text{reconstruction}} + \mathcal{L}_{\text{reward}} \tag{3.5}$$

3.1.2 Learning Architecture Configuration

The SRM utilizes distinct state representations for the critic and actor networks to address the asymmetry in available information during training and deployment. Table 3.1 summarizes the differences between the privileged observations (s_t^p) and the partially observable state (s_t^o) used as input to the SRM.

Observable(s_t^o) represents the information directly accessible from sensors in real-world scenarios, providing a partial view of the robot's state:

- *Joint positions*(q_i) *:* Represents the angular positions of the robot's joints, measured in radians (rad).
- *Joint velocities*($\dot{q}_i$)*:* Denotes the angular velocity of each joint, providing information on rotational speed in radians per second (rad/s).
- *Base angular velocity*($\omega_{x,y,z}$)*:* Measures the rotational speed of the robot's base along its axes, expressed in radians per second (rad/s).
- *Base Euler angles*($\phi_{x,y,z}$)*:* Describes the robot's orientation using roll, pitch, and yaw angles in radians (rad).
- *Previous actions*(a_t) *:* Tracks the actions executed in the previous time step to offer temporal context for decision-making.

SRM使用基于恢复的损失函数进行训练，该函数最小化恢复状态ˆs_t^f 与完整状态 s_t^f 之间的差异。恢复损失定义为：

$$\mathcal{L}_{\text{reconstruction}}(\hat{s}_t^f, s_t^f) = \begin{cases} \frac{1}{2}(\hat{s}_t^f - s_t^f)^2 & \text{if } |\hat{s}_t^f - s_t^f| \leq 1.0, \\ 1.0(|\hat{s}_t^f - s_t^f| - 0.5) & \text{otherwise} \end{cases} \tag{3.3}$$

确保重构状态在训练过程中紧密逼近评论家网络所使用的特权信息。

为了进一步提升任务特定性能，SRM引入了奖励一致性损失。该损失最小化使用完整状态 s_t^f 和重构状态ˆs_t^f 计算得到的奖励之间的差异，从而对齐两个状态中与任务相关的特征：

$$\mathcal{L}_{\text{reward}} = \|R(s_t^f, a_t) - R(\hat{s}_t^f, a_t)\|^2 \tag{3.4}$$

训练SRM的总损失表示为：

$$\mathcal{L}_{\text{SRM}} = \mathcal{L}_{\text{reconstruction}} + \mathcal{L}_{\text{reward}} \tag{3.5}$$

3.1.2 学习架构配置

SRM利用评论家网络和演员网络的不同状态表示，以解决训练和部署过程中可用信息的不对称性。表3.1总结了特权观测（s_t^p）与作为SRM输入的部分可观测状态（s_t^o）之间的差异。

可观测的(s_t^o) 表示在现实场景中可直接从传感器获取的信息，提供了机器人状态的部分视图：

- 关节位置(q_i)：表示机器人关节的角度位置，以弧度为单位进行测量（rad）。
- 关节速度($\dot{q}_i$): 表示每个关节的角速度，提供以弧度/秒为单位的旋转速度信息（rad/s）。
- 基座角速度($\omega_{x,y,z}$): 测量机器人基座沿其各轴的旋转速度，以弧度/秒为单位表示（rad/s）。
- 基座欧拉角($\phi_{x,y,z}$): 使用滚转角、俯仰角和偏航角（以弧度为单位）描述机器人的姿态（rad）。
- 先前动作(a_t)：追踪上一时间步执行的动作，以提供时间上下文用于决策。

- *Commands*$(cmd_{\dot{x},\dot{y},\omega_z})$: Represents the high-level velocity and directional commands provided to the robot.
- *History buffer*$(\mathcal{H}_k)$: Stores the states from time step $t-k$ to the current time step t, providing temporal context for the robot's decision-making process.

Privileged (s_t^p) is accessible only during simulation, this state provides richer information critical for policy optimization:

- *Base linear velocity*(v_x): Represents the forward velocity of the robot's base along the x-axis, measured in meters per second (m/s).
- *Current reward* (R_t): Indicates the immediate reward received by the robot, reflecting the effectiveness of the current actions.
- *Lift mask* (L_i): A boolean sensor data that indicates whether the robot has detected a curb or step, used to adjust hybrid locomotion.
- *Base height*(p_z): Specifies the vertical position of the robot's base relative to the ground, measured in meters (m).

By reconstructing $\hat{s}_t^f$ from s_t^o, the SRM aligns the state representations between the actor and critic, thereby reducing the disparity caused by the absence of privileged data. The reconstructed state $\hat{s}_t^f$ enables the actor network to make robust and consistent decisions in real-world environments. Additionally, while the SRM focuses on reconstructing the state, the role of the reward in shaping task-specific behaviors will be further detailed in Section 3.1.3. Finally, the complete SRM-PPO training procedure including rollout collection, SRM optimization (reconstruction and reward-consistency losses), and PPO policy/value updates is provided as pseudocode in Appendix 6.3.

- 指令$(cmd_{\dot{x},\dot{y},\omega_z})$: 表示提供给机器人的高级速度和方向指令。
- 历史缓冲区$(\mathcal{H}_k)$: 存储从时间步 $t-k$ 到当前时间步 t的状态，为机器人的决策过程提供时间上下文。

特权状态 (s_t^p) 仅在仿真期间可访问，该状态提供了对策略优化至关重要的更丰富信息:

- 基座线速度(v_x): 表示机器人基座沿x轴的前向速度，以米每秒(m/s)为单位测量。
- 当前奖励 (R_t): 指示机器人收到的即时奖励，反映了当前动作的有效性。
- 抬升掩码 (L_i): 一个布尔传感器数据，指示机器人是否检测到路缘或台阶，用于调整混合运动。
- 基座高度(p_z): 指定机器人基座相对于地面的垂直位置，以米(m)为单位测量。

通过从s_t^o重构$\hat{s}_t^f$，SRM对齐了演员网络与评论家网络之间的状态表示，从而减少了因缺乏特权数据而导致的差异。重构状态$\hat{s}_t^f$ 使演员网络能够在真实环境中做出稳健且一致的决策。此外，在SRM专注于重构状态的同时，奖励在塑造任务特定行为中的作用将在第3.1.3节中进一步详述。最后，完整的SRM-PPO训练流程（包括轨迹收集、SRM优化（重构损失和奖励一致性损失）以及PPO策略/价值更新）以伪代码形式提供于附录6.3。

Table 3.1: State Composition and Observability

State Component	Observable (s_t^o)	Privileged (s_t^p)
Joint position (q_i)	✓	
Joint velocity ($\dot{q}_i$)	✓	
Base angular velocity ($\omega_{x,y,z}$)	✓	
Base Euler angle ($\Theta_{x,y,z}$)	✓	
Previous actions (a_t)	✓	
Commands ($cmd_{\dot{x},\dot{y},\omega_z}$)	✓	
History buffer ($\mathcal{H}_k$)	✓	
Base linear velocity (v_x)		✓
Current reward (R_t)		✓
Lift mask (L_i)		✓
Base height (p_z)		✓

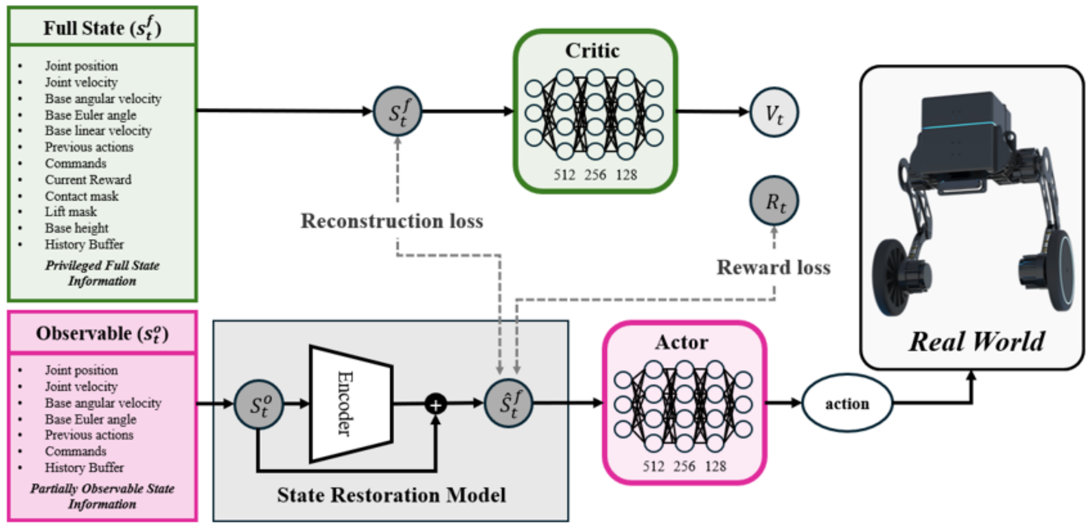

Figure 3.1: Overview of the State Restoration Model (SRM) architecture. The observable state s_t^o represents sensor-accessible data in real-world scenarios, including joint positions, velocities, angular velocity, Euler angles, previous actions, and commands. In simulation, the privileged full state s_t^f encompasses additional information such as contact forces, joint torques, base height, and environmental features. During training, the GRU-based encoder maps s_t^o to a reconstructed state $\hat{s}_t^f$, approximating s_t^f. The actor network operates on $\hat{s}_t^f$, leveraging enriched state information to bridge the sim-to-real gap and enable robust decision-making. The critic network, using s_t^f, evaluates the policy while the SRM minimizes both reconstruction loss and reward consistency loss to align $\hat{s}_t^f$ with s_t^f.

表 3.1：状态组成与可观测性

状态组件	可观测的 (s_t^o)	特权状态 (s_t^p)
关节位置 (q_i)	✓	
关节速度 ($\dot{q}_i$)	✓	
基座角速度（$\omega_{x,y,z}$）	✓	
基座欧拉角（$\Theta_{x,y,z}$）	✓	
先前动作（a_t）	✓	
指令（$cmd_{\dot{x},\dot{y},\omega_z}$）	✓	
历史缓冲区（$\mathcal{H}_k$）	✓	
基座线速度（v_x）		✓
当前奖励 (R_t)		✓
抬升掩码 (L_i)		✓
基座高度 (p_z)		✓

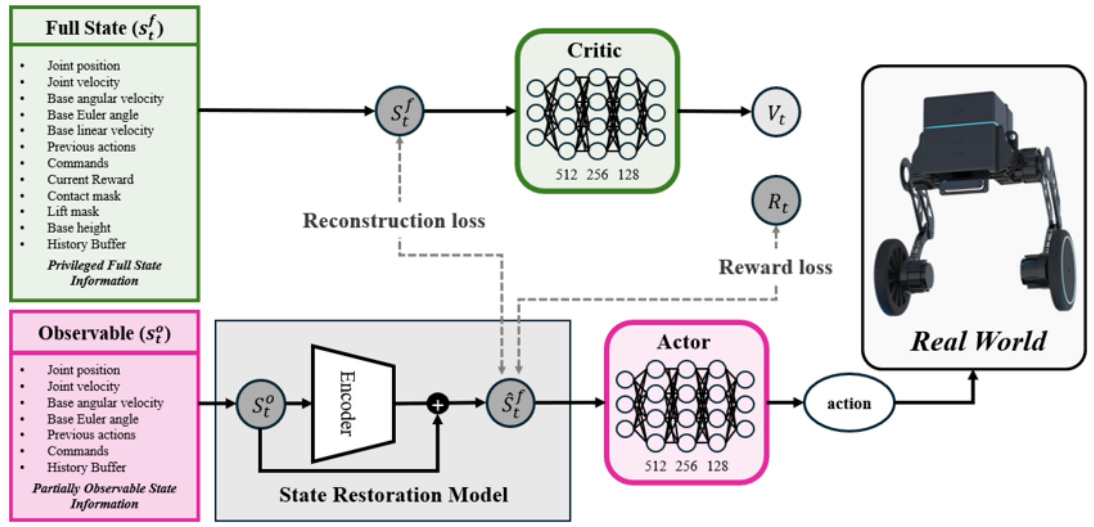

图 3.1：状态恢复模型 (SRM) 架构概览。可观测状态 s_t^o 代表现实世界场景中的传感器可访问数据，包括关节位置、速度、角速度、欧拉角、先前动作和指令。在仿真中，特权完整状态 s_t^f 包含额外信息，如接触力、关节力矩、基座高度和环境特征。在训练过程中，基于 GRU 的编码器将 s_t^o 映射到重构状态 $\hat{s}_t^f$，以近似 s_t^f。演员网络基于 $\hat{s}_t^f$ 运行，利用增强的状态信息来弥合仿真到现实差距，并实现稳健的决策。评论家网络使用 s_t^f 评估策略，同时 SRM 最小化重构损失和奖励一致性损失，以使 $\hat{s}_t^f$ 与 s_t^f 对齐。

3.1.3 Learning Hybrid Locomotion Policy

HLP is designed to enable seamless hybrid locomotion by dynamically integrating walking and rolling modes. The policy is implemented as a fully connected neural network comprising three hidden layers with 512, 256, and 128 ReLU-activated nodes. It processes the reconstructed full state ($\hat{s}_t^f$) generated by the SRM to output joint-level position and velocity targets ($q_1, q_2, \ldots, q_8$), facilitating robust and adaptive control.

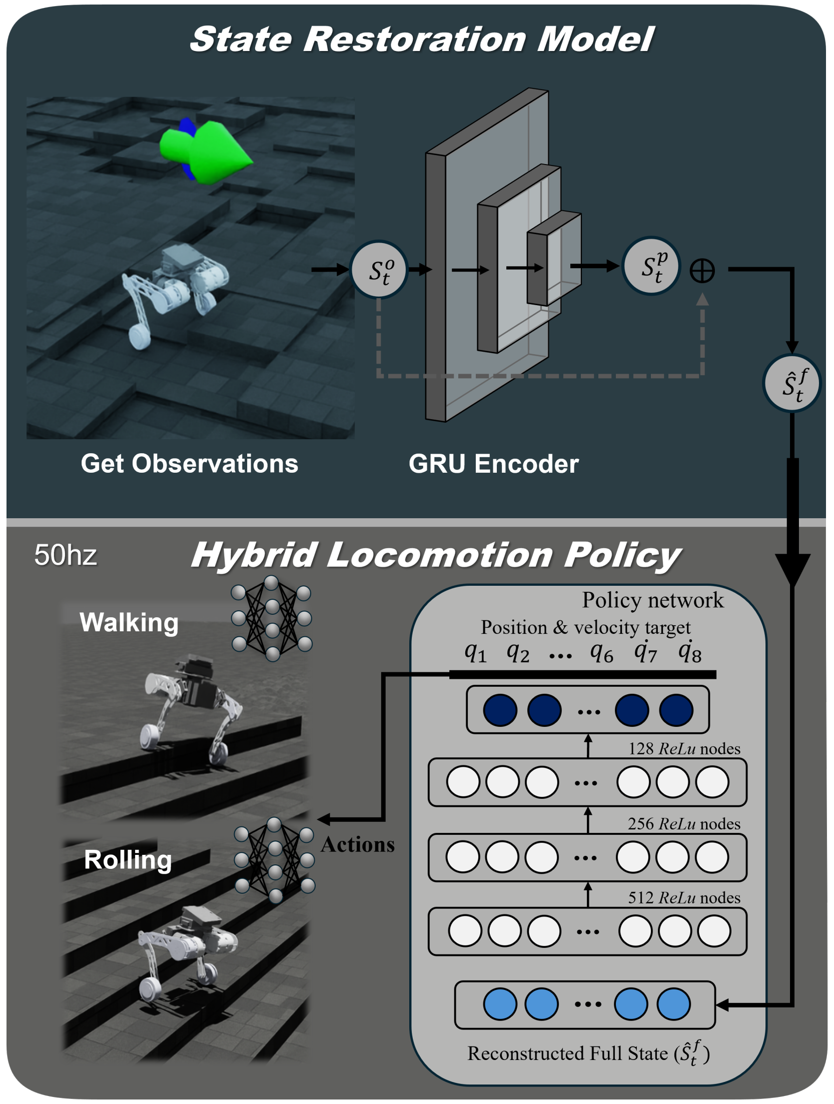

Figure 3.2: Overview of the Hybrid Locomotion Policy learning framework. The SRM reconstructs the full state ($\hat{s}_t^f$) from simulator observations and provides it as input to the Hybrid Locomotion Policy, which operates at 50 Hz. The policy network is a fully connected neural network comprising three hidden layers with 512, 256, and 128 ReLU-activated nodes. This network outputs joint-level position and velocity targets ($q_1, q_2, \ldots, \dot{q}_8, \dot{q}_8$), facilitating robust hybrid locomotion by dynamically switching between walking and rolling locomotion to adapt to varying terrains.

3.1.3 学习混合运动策略

HLP旨在通过动态整合行走与滚动模式，实现无缝的混合运动。该策略采用全连接神经网络实现，包含三个隐藏层，分别具有512、256和128个ReLU激活节点。它处理由SRM生成的<styled id='1'>^</styled>s_t^f 重构完整状态，以输出关节级位置和速度目标（q_1<styled id='5'>,</styled> q_2<styled id='8'>, . . . ,</styled> q_8），从而实现稳健且自适应的控制。

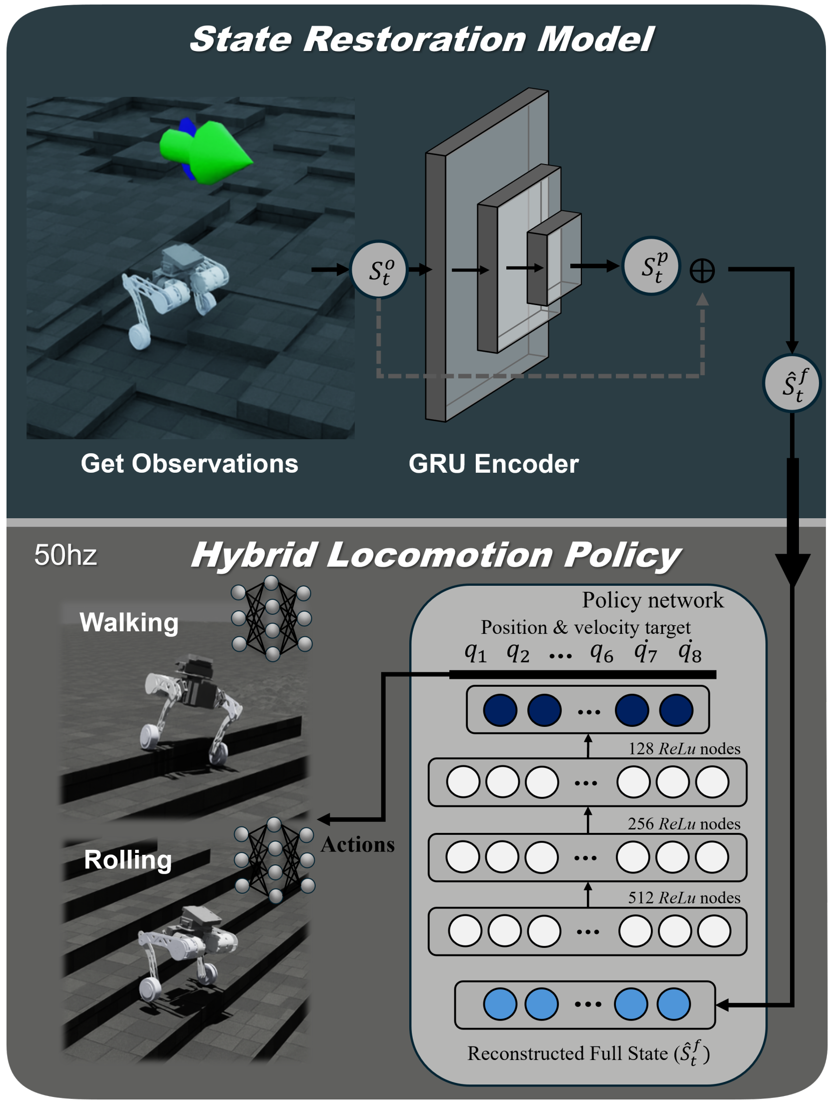

图3.2：混合运动策略学习框架概览。SRM从模拟器观测中重建完整状态（$\hat{}s_t^f$），并将其作为输入提供给以50赫兹运行的混合运动策略。策略网络是一个全连接神经网络，包含三个隐藏层，分别具有512、256和128个ReLU激活节点。该网络输出关节级位置和速度目标（$q_1, q_2, \ldots, \dot{q}_8$, $\dot{q}_8$），通过动态切换行走和滚动运动以适应变化地形，从而实现稳健的混合运动。

3.1.4 Hybrid Locomotion Policy Configuration

The Hybrid Locomotion Policy is configured to operate in both simulated and real-world environments by utilizing observations, applying domain randomization, and incorporating a structured reward mechanism. The key elements of the configuration are outlined as follows:

Observations and Domain Randomization

HLP utilizes observations derived from the robot's reconstructed full state, $\hat{s}_t^{f,H}$, as defined in the SRM architecture (see Section 3.1.2). A detailed breakdown of these observations, including their units and dimensions, is provided in Table 3.2.

To enhance robustness and adaptability in real-world conditions, domain randomization is applied during training. By introducing variability to observation parameters and environmental dynamics, the HLP is exposed to a wide range of randomized MDPs. This approach reduces overfitting to simulated environments and equips the policy to handle unseen scenarios during deployment. The domain randomization parameters, covering both the robot's internal states and external environmental factors, are summarized in Table 3.2.

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Reward Function

The reward function R_t is designed to optimize the HLP by addressing both task-specific objectives and regularization constraints. The total reward is expressed as:

$$\begin{aligned} R_t = & r^{track}_{t,cmd_{x,y}} + r^{track}_{t,cmd_z} + r^{reg}_{t,v_z} + r^{reg}_{t,\omega_{x,y}} \\ & + r^{reg}_{t,q_i} + r^{reg}_{t,\tau_i} + r^{reg}_{t,\dot{q}_i} + r^{reg}_{t,a_t} + r^{reg}_{t,p_z}, \end{aligned} \tag{3.6}$$

where $r^{track}_{t,cmd_{x,y}}$ tracks the desired linear velocity in the x and y directions, while r^{track}_{t,cmd_z} tracks the desired angular velocity around the z-axis for heading control. The regularization terms include r^{reg}_{t,v_z}, which penalizes vertical velocity to maintain stability, and $r^{reg}_{t,\omega_{x,y}}$, which penalizes angular velocities around the x and y axes to minimize unnecessary base rotations. Joint positions are regularized by r^{reg}_{t,q_i} to ensure they stay within allowable limits, while r^{reg}_{t,τ_i} penalizes excessive joint torques to promote energy efficiency. Further, $r^{reg}_{t,\dot{q}_i}$ penalizes joint accelerations to reduce mechanical strain, and r^{reg}_{t,a_t} regularizes

3.1.4 混合运动策略配置

混合运动策略被配置为通过利用观测数据、应用域随机化以及结合结构化奖励机制，在模拟环境和真实环境中运行。配置的关键要素概述如下：

观测与域随机化

HLP 利用从机器人重构完整状态中获得的观测值，$\hat{s}^{f}_{,t}{}^{H}$，如 SRM 架构中所定义（见第 3.1.2 节）。这些观测值的详细分解，包括其单位和维度，见表 3.2。

为了增强在真实世界条件下的鲁棒性和适应性，在训练过程中应用了域随机化。通过引入观测参数和环境动态的变异性，HLP 暴露于广泛的随机化马尔可夫决策过程中。这种方法减少了对模拟环境的过拟合，并使策略能够在部署期间处理未见过的场景。域随机化参数涵盖了机器人的内部状态和外部环境因素，总结于表 3.2。

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奖励函数

奖励函数 R_t 旨在通过处理任务特定目标和正则化约束来优化 HLP。总奖励表示为：

$$\begin{aligned} R_t = & r^{track}_{t,cmd_{x,y}} + r^{track}_{t,cmd_z} + r^{reg}_{t,v_z} + r^{reg}_{t,\omega_{x,y}} \\ & + r^{reg}_{t,q_i} + r^{reg}_{t,\tau_i} + r^{reg}_{t,\dot{q}_i} + r^{reg}_{t,a_t} + r^{reg}_{t,p_z}, \end{aligned} \tag{3.6}$$

其中 $r^{track}_{t}{}_{,cmd_{x,y}}$ 追踪 x 和 y 方向上的期望线速度，而 $r^{track}_{t}{}_{,cmd_z}$ 追踪围绕 z 轴的期望角速度以实现航向控制。正则化项包括 $r^{reg}_{t}{}_{,v_z}$，它惩罚垂直速度以维持稳定性，以及 $r^{reg}_{t}{}_{,\omega_{x,y}}$，它惩罚围绕 x 和 y 轴的角速度以最小化不必要的基座旋转。关节位置由 $r^{reg}_{t}{}_{,q_i}$ 进行正则化，以确保它们保持在允许范围内，而 $r^{reg}_{t}{}_{,\tau_i}$ 惩罚过大的关节力矩以提升能效。此外，$r^{reg}_{t\,\dot{q}_i}$，惩罚关节加速度以减少机械应变，并且 $r^{reg}_{t}{}_{,a_t}$ 正则化

Table 3.2: Domain randomization parameters and configuration for HLP

Parameters	Dim	Unit	Range	Operator
q_i	6	rad	$[-0.04, 0.04]$	additive
$\dot{q}_i$	8	rad/s	$[-1.0, 1.0]$	additive
$\phi_{x,y,z}$	3	rad	$[-0.125, 0.125]$	additive
$\omega_{x,y,z}$	3	rad/s	$[-1.0, 1.0]$	additive
$v_{x,y,z}$	3	rad/s	$[-0.15, 0.15]$	additive
p_z	1	m	$[-0.05, 0.05]$	additive
τ_i	8	Nm	$[-0.25, 0.25]$	additive
$\ddot{q}_i$	8	rad/s^2	$[-1.75, 1.75]$	additive
H_t	64	m	$[-0.025, 0.025]$	additive
$cmd_{x,y,z}$	3	m/s	$[-2.0, 2.0]$	additive
$delay_{system}$	-	ms	$[0.0, 20.0]$	additive
$friction_{static}$	-	μ	$[0.3, 1.0]$	scaling
$friction_{dynamic}$	-	μ	$[0.6, 1.0]$	scaling
$mass_{body}$	-	kg	$[-1.25, 3.5]$	additive
$push$	-	s	$[10.0, 15.0]$	additive

表3.2：HLP的域随机化参数与配置

参数	Dim	Unit	范围	算子
q_i	6	rad	$[-0.04, 0.04]$	加性
$\dot{q}_i$	8	弧度/秒	$[-1.0, 1.0]$	加性
$\phi_{x,y,z}$	3	rad	$[-0.125, 0.125]$	加性
$\omega_{x\ y,z}$	3	弧度/秒	$[-1.0, 1.0]$	加性
$v_{x\ y,z}$	3	弧度/秒	$[-0.15, 0.15]$	加性
p_z	1	m	$[-0.05, 0.05]$	加性
τ_i	8	Nm	$[-0.25, 0.25]$	加性
$\ddot{q}_i$	8	弧度/s^2	$[-1.75, 1.75]$	加性
H_t	64	m	$[-0.025, 0.025]$	加性
$cmd_{x,y,z}$	3	m/s	$[-2.0, 2.0]$	加性
$delay_{system}$	-	ms	$[0.0, 20.0]$	加性
$friction_{static}$	-	μ	$[0.3, 1.0]$	缩放
$friction_{dynamic}$	-	μ	$[0.6, 1.0]$	缩放
$mass_{body}$	-	kg	$[-1.25, 3.5]$	加性
$push$	-	s	$[10.0, 15.0]$	加性

– 12 –

the rate of change in actions to ensure smooth control inputs. Finally, r^{reg}_{t,p_z} encourages the robot's base height to remain within a stable range for better terrain adaptability.

The detailed mathematical definitions of these reward components are provided in Table 3.3.

Table 3.3: Reward Functions and Weights for the Hybrid Locomotion Policy

Reward	**Equation** (r_t)	**Weight** (w)	**Params** (ξ)
$r^{track}_{t,cmd_{x,y}}$	$w \cdot \exp\left(-\xi \cdot \|cmd_{x,y} - v_{x,y}\|^2\right)$	2.0	4.0
$r^{track}_{t,cmd_{\omega_z}}$	$w \cdot \exp(-\xi \cdot \|cmd_{\omega_z} - \omega_z)\|^2)$	1.0	4.0
r^{reg}_{t,v_z}	$-w \cdot \|v_z\|^2$	0.1	-
$r^{reg}_{t,\omega_{x,y}}$	$-w \cdot \|\omega_{x,y}\|^2$	0.05	-
r^{reg}_{t,q_i}	$-w \cdot \|q_{2i} - \xi\|^2, i \in \{0,1,2,3\}$	0.2	q_{2i+1}
r^{reg}_{t,q_i}	$-w \cdot \|q_i - \xi\|^2$	0.1	target
r^{reg}_{t,τ_i}	$-w \cdot \|\tau_i\|^2$	0.05	-
$r^{reg}_{t,\dot{q}_i}$	$-w \cdot \|\dot{q}_i\|^2$	0.001	-
r^{reg}_{t,a_t}	$-w \cdot \|a_t - a_{t-1}\|^2$	$5.0 \cdot 10^{-4}$	-
r^{reg}_{t,p_z}	$-w \cdot \|p_z - \xi\|^2$	20.0	0.40187

动作变化率以确保平滑控制输入。最后，r_t^{reg} $_{,p_z}$ 鼓励机器人的基座高度保持在稳定范围内，以获得更好的地形适应性。

这些奖励组件的详细数学定义见表 e 3.3。

表 3.3：混合运动策略的奖励函数与权重

奖励	**公式** (r_t)	**权重** (w)	**参数** (ξ)
$r^{track}_{t,cmd_{x,y}}$	$w \cdot \exp\left(-\xi \cdot \|cmd_{x,y} - v_{x,y}\|^2\right)$	2.0	4.0
$r^{track}_{t,cmd_{\omega_z}}$	$w \cdot \exp(-\xi \cdot \|cmd_{\omega_z} - \omega_z)\|^2)$	1.0	4.0
r^{reg}_{t,v_z}	$-w \cdot \|v_z\|^2$	0.1	-
$r^{reg}_{t,\omega_{x,y}}$	$-w \cdot \|\omega_{x\ y}\|^2$	0.05	-
r^{reg}_{t,q_i}	$-w \cdot \|q_{2i} - \xi\|^2, i \in \{0,1,2,3\}$	0.2	q_{2i+1}
r^{reg}_{t,q_i}	$-w \cdot \|q_i - \xi\|^2$	0.1	目标
r^{reg}_{t,τ_i}	$-w \cdot \|\tau_i\|^2$	0.05	-
$r^{reg}_{t\ \dot{q}_i,}$	$-w \cdot \|\dot{q}_i\|^2$	0.001	-
r^{reg}_{t,a_t}	$-w \cdot \|a_t - a_{t-1}\|^2$	$5.0 \cdot 10^{-4}$	-
r^{reg}_{t,p_z}	$-w \cdot \|p_z - \xi\|^2$	20.0	0.40187

IV. Experiment

4.1 Experimental Setup

This section describes the evaluation pipeline and experimental settings used to validate the proposed SRM-based framework across (i) simulation training, (ii) cross-simulator transfer, and (iii) real-world deployment. Concretely, we follow a *simulation* → *simulation* → *real* procedure, where the policy is trained in a high-throughput simulator and then validated in an independent physics engine before deployment on the physical robot (Fig. 4.1). For simulation, we use Isaac Lab built on NVIDIA Isaac Sim [17] and evaluate transfer in MuJoCo [18].

4.1.1 Simulation → Simulation → Real Pipeline

Stage 1: Training in simulation. Training is performed in simulation to enable fast, safe, and scalable rollout generation. We train the locomotion policy using PPO [9] in Isaac Lab [17]; the key network architectures and optimization hyperparameters are summarized in Appendix 6.1. During training, we monitor learning progress using (i) the mean episodic reward and (ii) task-specific tracking reward terms (e.g., forward-velocity tracking). We additionally conduct an ablation by training two policies under identical settings: *with SRM* and *without SRM*. This allows us to compare learning dynamics (e.g., reward growth rate and final reward at a fixed number of steps) and quantify whether SRM improves optimization under partial observability.

Stage 2: Sim-to-sim validation. Before hardware deployment, the trained policy is evaluated in MuJoCo [18] to reduce simulator-specific overfitting and to verify that learned behaviors reproduce under a different physics stack (Fig. 4.1). This stage provides a controlled validation step for velocity tracking and stability tests prior to real-world experiments. In particular, we evaluate step-command tracking over a range of commanded forward velocities and compare SRM versus the baseline under identical command profiles.

Stage 3: Sim-to-real deployment. Finally, the policy is transferred to the physical robot and evaluated in real-world trials, focusing on commanded-velocity tracking and stability under practical sensing constraints.

IV. 实验

4.1 实验设置

本节描述了用于验证所提出的基于SRM的框架的评估流程和实验设置，涵盖以下三个方面：(i) 仿真训练，(ii) 跨仿真器迁移，以及(iii) 真实世界部署。具体而言，我们遵循仿真 → 仿真 → 真实 流程，即策略在高通量仿真器中进行训练，然后在独立的物理引擎中验证，最后部署到物理机器人上（图4.1）。在仿真方面，我们使用基于NVIDIA Isaac Sim[17]构建的Isaac Lab，并在MuJoCo [18]中评估迁移效果。

4.1.1 仿真 →仿真 →真实管道

阶段 1：在仿真中训练。训练在仿真环境中进行，以实现快速、安全且可扩展的轨迹生成。我们使用 PPO [9]在 Isaac Lab [17]中训练运动策略；关键的网络架构和优化超参数总结于附录6.1。训练期间，我们通过 (i) 平均回合奖励和 (ii) 任务特定跟踪奖励项（例如，前向速度跟踪）来监控学习进度。此外，我们进行了一项消融实验，在相同设置下训练两个策略：使用 *SRM* 和不使用 *SRM*。这使我们能够比较学习动态（例如，奖励增长率和固定步数下的最终奖励），并量化 SRM 是否在部分可观测性下改善了优化。

阶段 2：仿真到仿真验证。在硬件部署之前，训练好的策略在 MuJoCo [18]中进行评估，以减少仿真器特定过拟合，并验证学习到的行为是否能在不同的物理引擎下复现（图 4.1）。此阶段为真实世界实验前的速度跟踪和稳定性测试提供了一个受控的验证步骤。具体来说，我们评估了在一系列指令前向速度范围内的步进指令跟踪，并在相同的指令配置文件下比较了 SRM 与基线。

阶段三：仿真到真实部署。最后，将策略迁移至物理机器人，并在真实世界试验中评估其指令速度跟踪能力及实际感知约束下的稳定性。

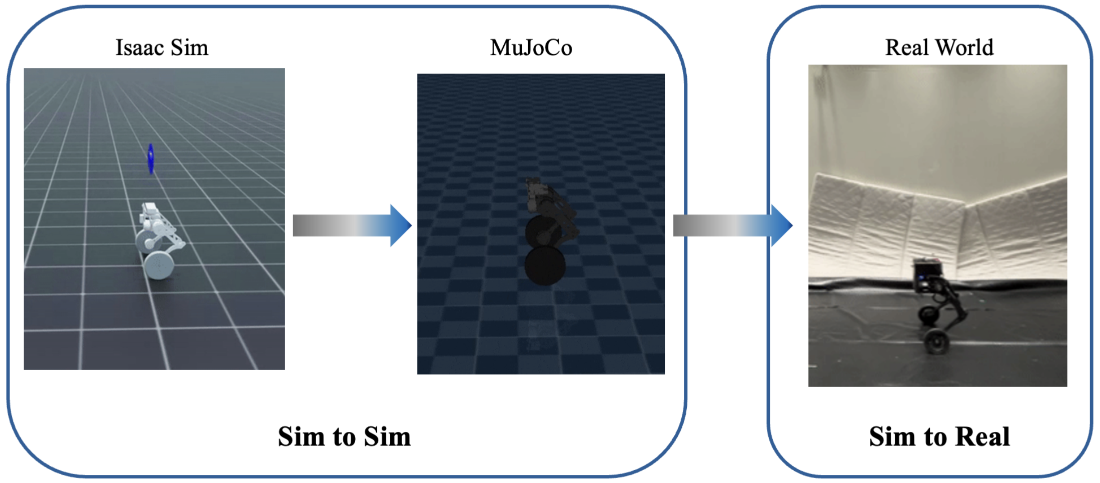

Figure 4.1: Evaluation pipeline: training in simulation (Isaac Sim wth Isaac Lab), cross-simulator validation (MuJoCo), and final deployment on the real robot.

4.1.2 Real-World Measurement Setup

To quantify tracking performance on hardware, we estimate the robot's linear velocity from external motion-capture measurements. Specifically, linear velocity is computed from OptiTrack marker positions using a first-order forward Euler difference:

$$\mathbf{v}(t_k) \approx \frac{\mathbf{p}(t_k) - \mathbf{p}(t_{k-1})}{\Delta t}, \quad \Delta t = t_k - t_{k-1}. \tag{4.1}$$

This procedure corresponds to the explicit (forward) Euler method for first-order numerical differentiation integration [19]. In our setup, the resulting velocity estimate has an approximate measurement error of $0.06\ \mathrm{m/s}$.

4.1.3 Evaluation Protocol

Across both simulators and real-world experiments, we assess the controller primarily through commanded-velocity tracking tests using step-command scenarios. Each trial consists of (i) a commanded forward velocity profile and (ii) the measured forward velocity response. We compute tracking errors after time alignment between the command and measured signals.

Command profiles. We evaluate step commands spanning a representative range of forward velocities (e.g., 0.5–$1.0\ \mathrm{m/s}$ in simulation) to test both transient response and steady-state regulation. The same class of step inputs is used for cross-simulator validation and real-world trials to keep evaluation conditions consistent across the pipeline.

Metrics. For each trial, we report time-domain tracking errors over a fixed evaluation horizon:

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{k=1}^{T}\left(v_x(t_k) - v_x^{\mathrm{cmd}}(t_k)\right)^2}, \quad \mathrm{MAE} = \frac{1}{T}\sum_{k=1}^{T}\left|v_x(t_k) - v_x^{\mathrm{cmd}}(t_k)\right|. \tag{4.2}$$

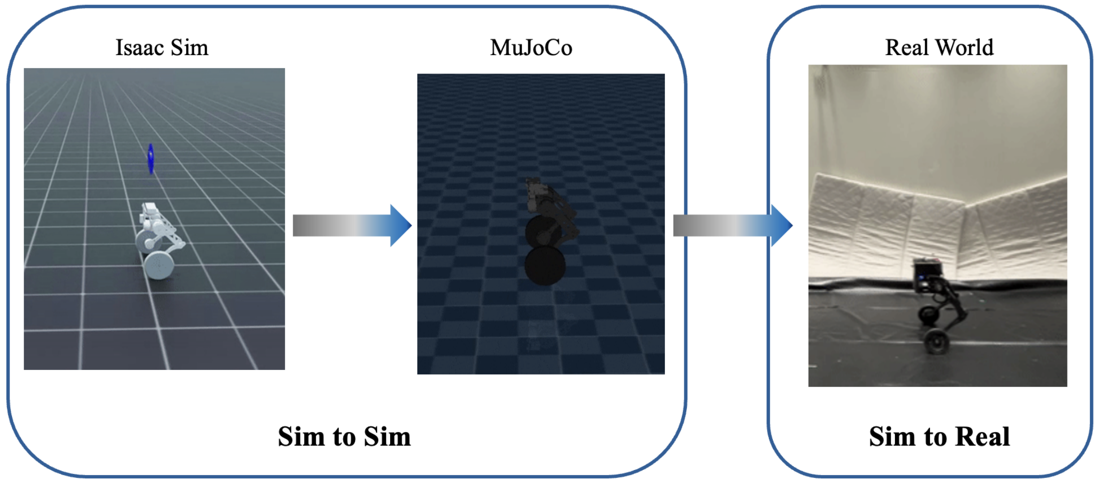

图 4.1：评估流程：在仿真中训练（使用 Isaac Lab 的 Isaac Sim），跨仿真器验证（MuJoCo），以及在真实机器人上的最终部署。

4.1.2 真实世界测量设置

为量化硬件上的跟踪性能，我们通过外部动作捕捉测量来估算机器人的线速度。具体而言，线速度是根据OptiTrack标记点位置，采用一阶前向欧拉差分计算得出：

$$\mathbf{v}(t_k) \approx \frac{\mathbf{p}(t_k) - \mathbf{p}(t_{k-1})}{\Delta t}, \quad \Delta t = t_k - t_{k-1}. \tag{4.1}$$

该过程对应于一阶数值微分积分的显式（前向）欧拉法 [19]。在我们的设置中，由此产生的速度估计具有约 $0.06\ \mathrm{m/s}$ 的测量误差。

4.1.3 评估协议

在仿真器和真实世界实验中，我们主要通过使用阶跃指令场景的指令速度跟踪测试来评估控制器。每次试验包括：(i) 一个指令前向速度配置文件，以及 (ii) 测量得到的前向速度响应。我们在指令信号与测量信号之间进行时间对齐后计算跟踪误差。

指令配置文件。我们评估覆盖代表性前向速度范围（例如，仿真中 0.5–$1.0\mathrm{m/s}$ ）的阶跃指令，以测试瞬态响应和稳态调节。相同的阶跃输入类别用于跨仿真器验证和真实世界试验，以保持整个评估流程中评估条件的一致性。

M指标。 对于每次试验，我们在固定的评估时域内报告时域跟踪误差：

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{k=1}^{T}\left(v_x(t_k) - v_x^{\mathrm{cmd}}(t_k)\right)^2}, \quad \mathrm{MAE} = \frac{1}{T}\sum_{k=1}^{T}\left|v_x(t_k) - v_x^{\mathrm{cmd}}(t_k)\right|. \tag{4.2}$$

To separate transient and steady-state behavior for step responses, we additionally compute *steady-state* metrics over a terminal window of fixed duration (e.g., the last 2 s of a trial), denoted as w_RMSE and w_MAE. This windowed evaluation reduces sensitivity to initial transients and emphasizes final tracking accuracy under constant command.

4.2 Experimental Results

We report experimental results following the *simulation* → *simulation* → *real* evaluation pipeline. Unless otherwise stated, we compare two policies trained under identical settings: a baseline trained *without SRM* and the proposed method trained *with SRM*. We evaluate (i) training performance, (ii) velocity tracking performance under simulation-to-simulation transfer (Isaac Sim → MuJoCo), and (iii) velocity tracking performance on the real robot.

4.2.1 Training Performance Comparison

To analyze the impact of SRM on policy learning, we compare the reward progression of policies trained *with* and *without* SRM under identical training settings. Figure 4.2 shows the evolution of the mean episodic reward and the tracking reward for forward linear velocity (v_x) over 10,000 training steps.

At 10,000 steps, the policy trained with SRM achieves a higher mean reward (48.03) than the policy trained without SRM (45.19). In addition, the v_x tracking reward is slightly higher with SRM (1.8347) than without SRM (1.8037). These results are consistent with SRM providing a more informative state-like representation from partial observations, which improves optimization and tracking-related reward terms.

4.2.2 Simulation-to-Simulation Results (Isaac Sim → MuJoCo)

We next report *simulation-to-simulation* results to assess cross-simulator robustness by executing the learned policy in an independent physics engine (MuJoCo). This evaluation reduces the risk of simulator-specific overfitting and focuses on commanded-velocity tracking under step inputs.

Velocity Tracking Performance

To quantitatively evaluate forward velocity tracking in the sim-to-sim setting, we conduct step-command tests at three commanded speeds (0.5, 0.75, and 1.0 m/s). We report the settling time and steady-state error in Table 4.1. Settling time is defined as the time required for the base forward velocity to remain within ±5% of the commanded velocity. Steady-state error is defined as the absolute difference between the steady-state base velocity and the commanded velocity.

为了区分阶跃响应的瞬态和稳态行为，我们额外在固定持续时间的终端窗口（例如，试验的最后 2 秒）上计算 稳态指标，记为 w 均方根误差和w 平均绝对误差。这种窗口评估降低了对初始瞬态的敏感性，并强调了恒定指令下的最终跟踪精度。

4.2 实验结果

我们按照仿真 → 仿真 → 真实评估流程报告实验结果。除非另有说明，我们比较在相同设置下训练的两个策略：一个是在无*SRM*情况下训练的基线，另一个是在有*SRM*情况下训练的提出方法。我们评估：(i) 训练性能，(ii) 仿真到仿真迁移（Isaac Sim → MuJoCo）下的速度跟踪性能，以及 (iii) 在真实机器人上的速度跟踪性能。

4.2.1 训练性能比较

为了分析SRM对策略学习的影响，我们比较了在相同训练设置下，有和无SRM训练的策略的奖励进展。图4.2展示了在10,000 训练步数内，平均回合奖励和前向线速度（v_x）的跟踪奖励的演变。

在10,000 步时，使用SRM训练的策略获得了更高的平均奖励（48.03），而未经SRM训练的策略为45.19。此外，v_x 跟踪奖励在有SRM（1.8347）的情况下略高于无SRM（1.8037）。这些结果与SRM从部分观测中提供更具信息性的状态类表示相一致，从而改善了优化和与跟踪相关的奖励项。

4.2.2 仿真到仿真结果（Isaac Sim → MuJoCo）

接下来，我们报告仿真到仿真的结果，通过在独立的物理引擎（MuJoCo）中执行学习策略来评估跨仿真器鲁棒性。该评估降低了仿真器特定过拟合的风险，并专注于阶跃输入下的指令速度跟踪。

速度跟踪性能

为了在仿真到仿真设置中定量评估前向速度跟踪，我们在三个指令速度（0.5、0.75和1.0 m/s）下进行了阶跃指令测试。我们在表4.1中报告了稳定时间和稳态误差。稳定时间定义为机体前向速度保持在指令速度的 ±5% 范围内所需的时间。稳态误差定义为稳态机体速度与指令速度之间的绝对差值。

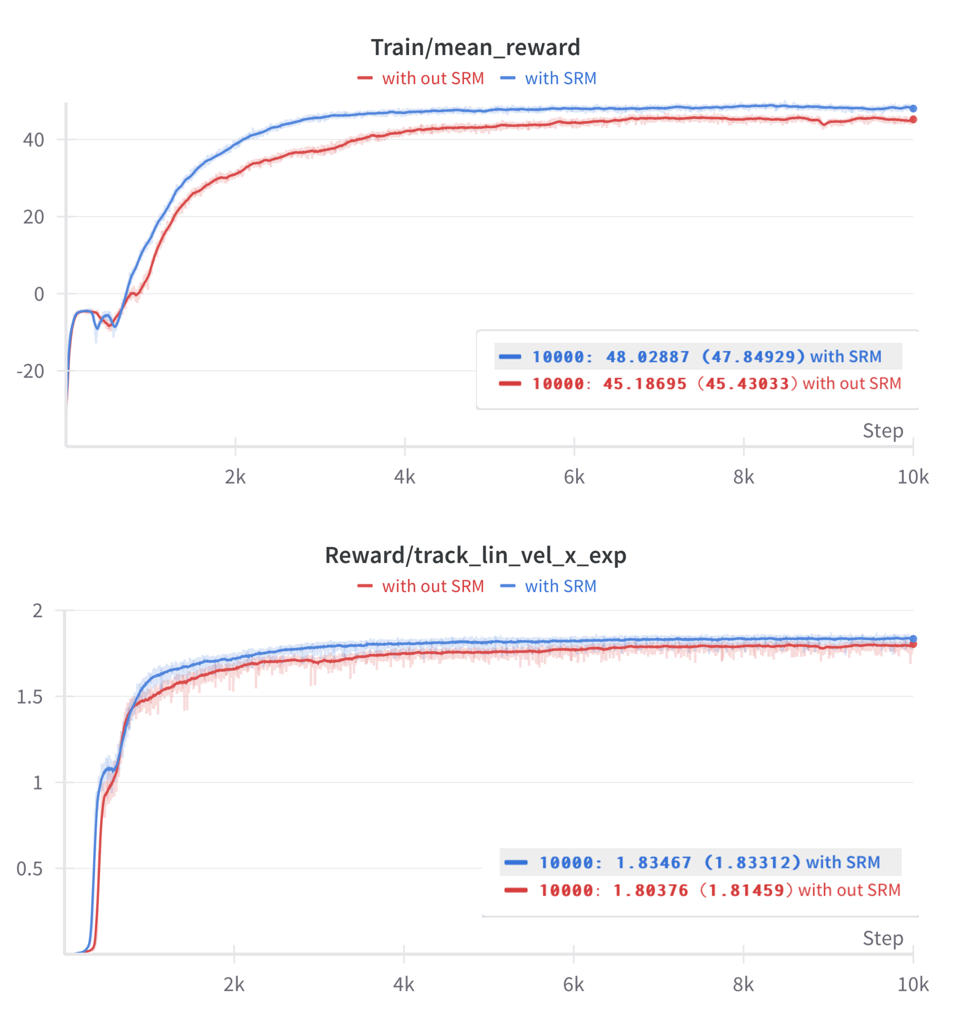

Figure 4.2: Training performance comparison of policies with and without SRM. The top plot shows the mean episodic reward, and the bottom plot shows the forward velocity tracking reward (v_x).

Table 4.1: Velocity tracking performance for step input commands (sim-to-sim).

Command (m/s)	Condition	Settling Time (s)	Steady-State Error (m/s)
0.50	with SRM	**0.62**	**0.0002**
	without SRM	0.50	0.1058
0.75	with SRM	**0.68**	**0.0020**
	without SRM	0.60	0.0771
1.00	with SRM	**0.76**	**0.0039**
	without SRM	0.74	0.0295

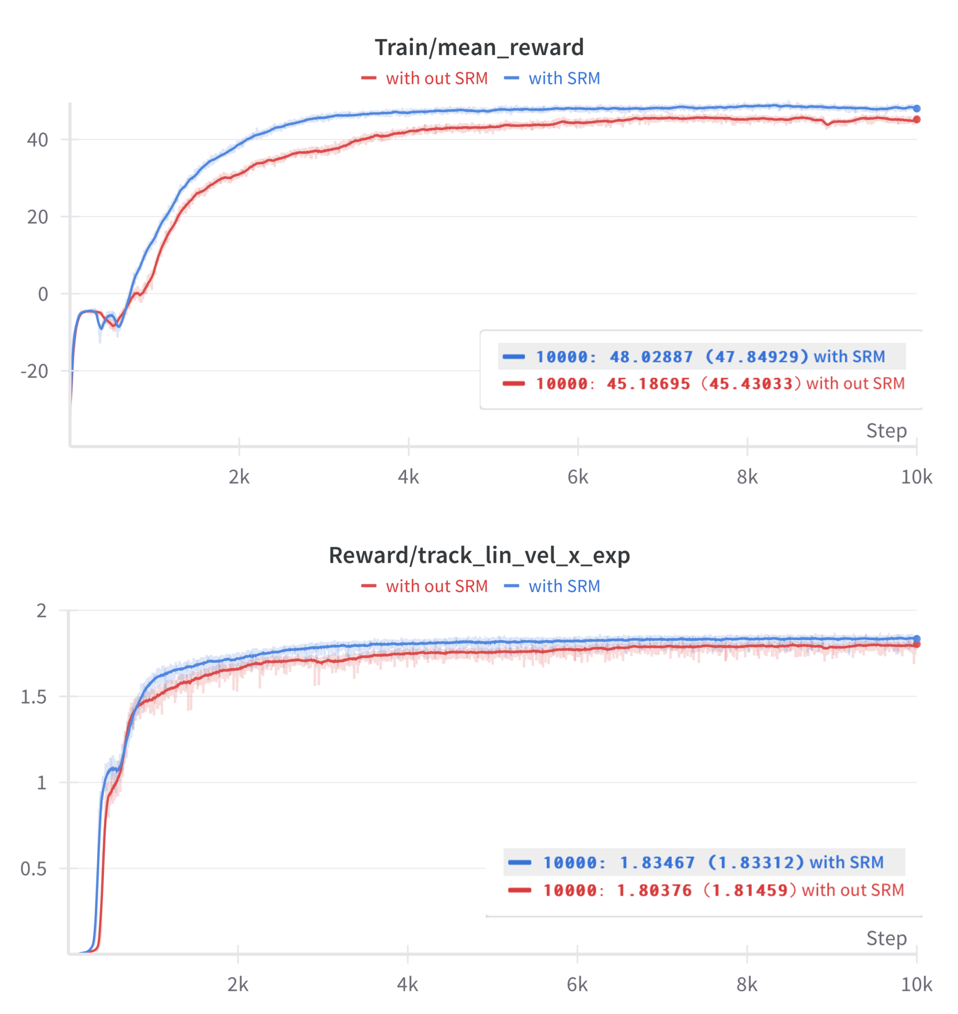
Train/mean_reward
with out SRM
with SRM
10000: 48.02887 (47.84929) with SRM
10000: 45.18695 (45.43033) with out SRM
Step
Reward/track_lin_vel_x_exp
with out SRM
with SRM
10000: 1.83467 (1.83312) with SRM
10000: 1.80376 (1.81459) with out SRM
Step

图4.2：有无SRM策略的训练性能对比。上图显示平均回合奖励，下图显示前向速度跟踪奖励（v_x）。

表4.1：阶跃输入指令的速度跟踪性能（仿真到仿真）。

指令（米/秒）	条件	稳定时间（秒）	稳态误差（米/秒）
0.50	使用 SRM	**0.62**	**0.0002**
	不使用 SRM	0.50	0.1058
0.75	使用 SRM	**0.68**	**0.0020**
	不使用 SRM	0.60	0.0771
1.00	使用 SRM	**0.76**	**0.0039**
	不使用 SRM	0.74	0.0295

Across all tested speeds, SRM substantially reduces steady-state error. For example, at a commanded velocity of $0.5\ \mathrm{m/s}$, the steady-state error decreases from $0.1058\ \mathrm{m/s}$ to $0.0002\ \mathrm{m/s}$. At $1.0\ \mathrm{m/s}$, the error decreases from $0.0295\ \mathrm{m/s}$ to $0.0039\ \mathrm{m/s}$. Settling time increases slightly with SRM in these tests, indicating a more conservative transient response while achieving lower steady-state bias.

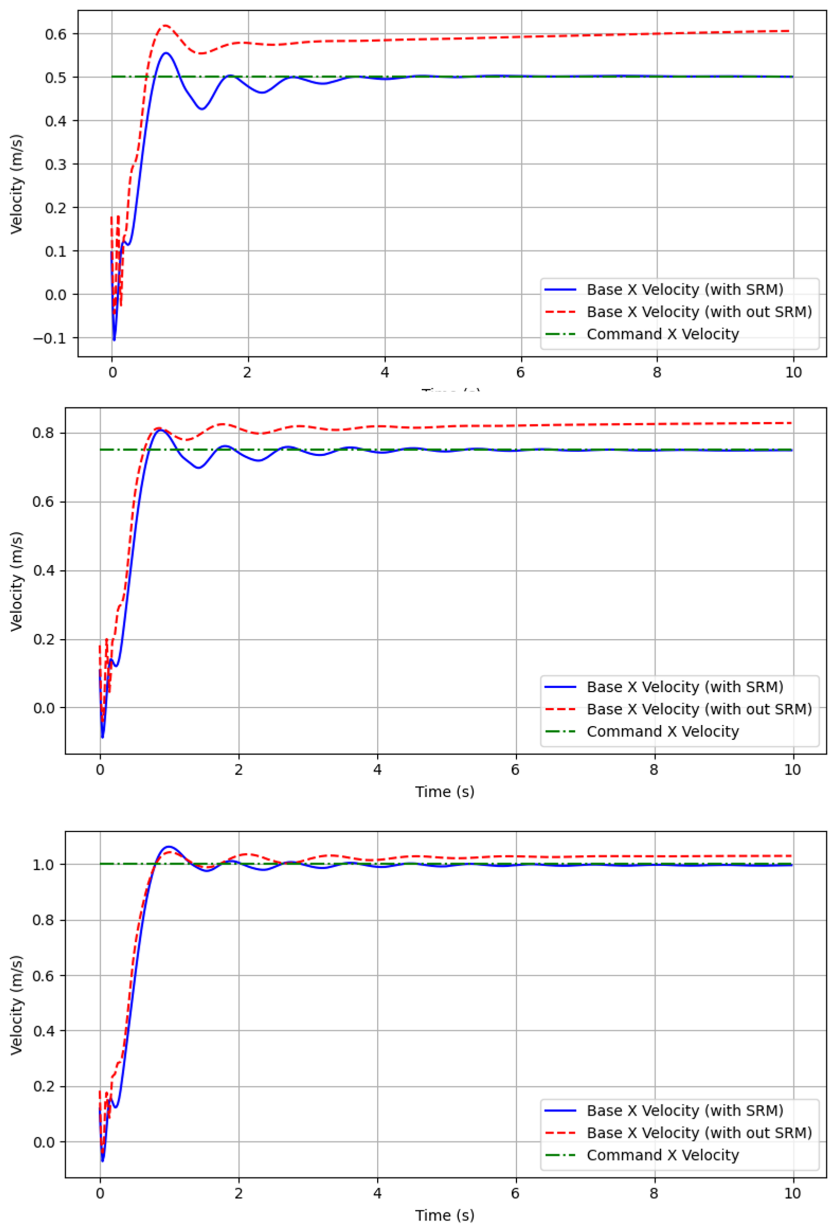

Figure 4.3: Forward velocity response to step input commands with and without SRM (sim-to-sim), showing reduced steady-state error with SRM.

在所有测试速度下，SRM 显著降低了稳态误差。例如，在指令速度为 $0.5\ \mathrm{m/s}$ 时，稳态误差从 $0.1058\ \mathrm{m/s}$ 降低到 $0.0002\ \mathrm{m/s}$。在 $1.0\ \mathrm{m/s}$ 时，误差从 $0.0295\ \mathrm{m/s}$ 降低到 $0.0039\ \mathrm{m/s}$。在这些测试中，SRM 的稳定时间略有增加，表明在实现更低稳态偏差的同时，瞬态响应更为保守。

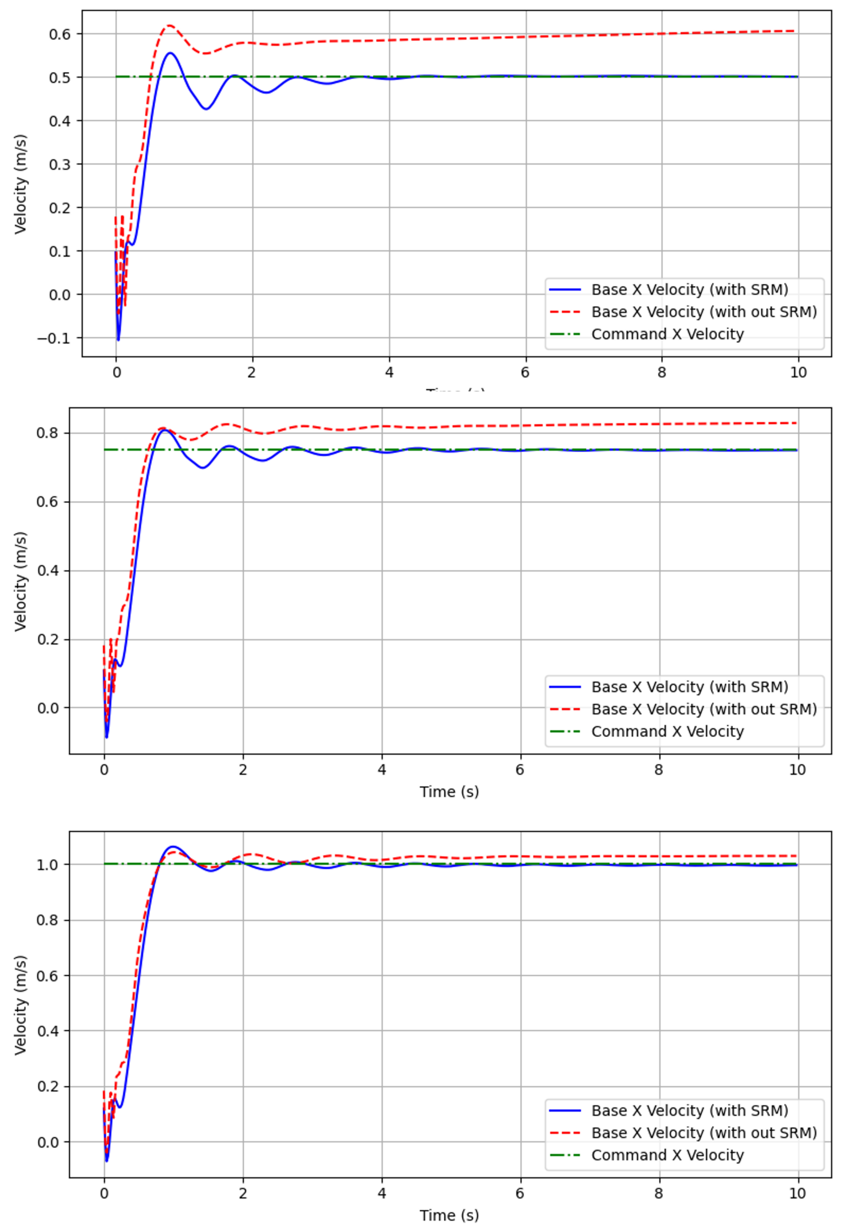

图4.3：有无SRM（仿真到仿真）情况下对阶跃输入指令的前向速度响应，显示使用SRM后稳态误差减小。

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Table 4.2: Real-world forward-velocity tracking errors.

Condition	RMSE	MAE	w_{RMSE}	w_{MAE}
with SRM **(proposed)**	**0.291831**	**0.230260**	**0.109146**	**0.090787**
without SRM (baseline)	0.673222	0.649399	0.631776	0.596709

4.2.3 Real-World Velocity Tracking Performance

We evaluate sim-to-real deployment by running the learned policy on the physical robot and estimating forward velocity using the OptiTrack-based procedure described in Section 4.1.2. We report tracking behavior by comparing the commanded forward velocity v_x^{cmd} and the measured response v_x, and summarize performance using standard tracking-error metrics. Full time-series plots for additional trials are provided in Appendix 6.2.

Quantitative comparison

Table 4.2 summarizes real-world tracking errors for the proposed SRM policy and a baseline trained without SRM. Because it is difficult to reproduce perfectly identical command trajectories across hardware runs, our comparison is conducted under the same evaluation protocol (time alignment, evaluation horizon, and metric definitions), rather than requiring trial-wise identical command sequences.

Over the evaluation horizon, SRM reduces RMSE from 0.673222 to **0.291831** and MAE from 0.649399 to **0.230260**. In addition, steady-state tracking performance over the last 2 s improves substantially: w_{RMSE} decreases from 0.631776 to **0.109146** and w_{MAE} decreases from 0.596709 to **0.090787**. These results indicate that SRM improves both overall tracking accuracy and final regulation behavior on hardware.

Qualitative behavior

Figure 4.4 shows representative real-world trials for the SRM-enabled policy and the baseline, including aligned command/response trajectories and the associated error summaries. Although the exact command sequences differ between hardware runs, both trials demonstrate typical step-like command changes and highlight the qualitative difference in tracking behavior. The SRM-enabled policy exhibits smaller tracking deviations and reduced steady-state bias in the presented trial, which is consistent with the quantitative comparison in Table 4.2. Additional real-world plots are provided in Appendix 6.2.

表4.2：真实世界前向速度跟踪误差。

条件	RMSE	MAE	w_{RMSE}	w_{MAE}
使用SRM **（提出的）**	**0.291831**	**0.230260**	**0.109146**	**0.090787**
不使用SRM（基线）	0.673222	0.649399	0.631776	0.596709

4.2.3 真实世界速度跟踪性能

我们通过在第 4.1.2 节描述的基于 OptiTrack 的程序，在物理机器人上运行学习策略并估计前向速度，来评估仿真到真实部署。我们通过比较指令前向速度 v_x^{cmd} 和测量响应 v_x 来报告跟踪行为，并使用标准跟踪误差指标总结性能。其他试验的完整时间序列图见附录 6.2。

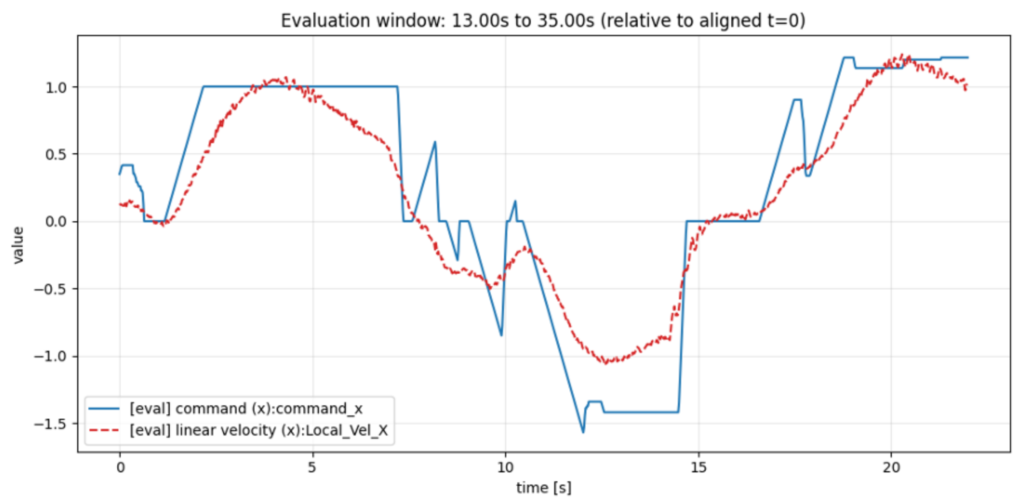
定量比较

表4.2总结了所提出的SRM策略与未使用SRM训练的基线的实际世界跟踪误差。由于在硬件运行中难以复现完全相同的命令轨迹，我们的比较是在相同的评估协议（时间对齐、评估时域和指标定义）下进行的，而非要求每次试验使用相同的命令序列。

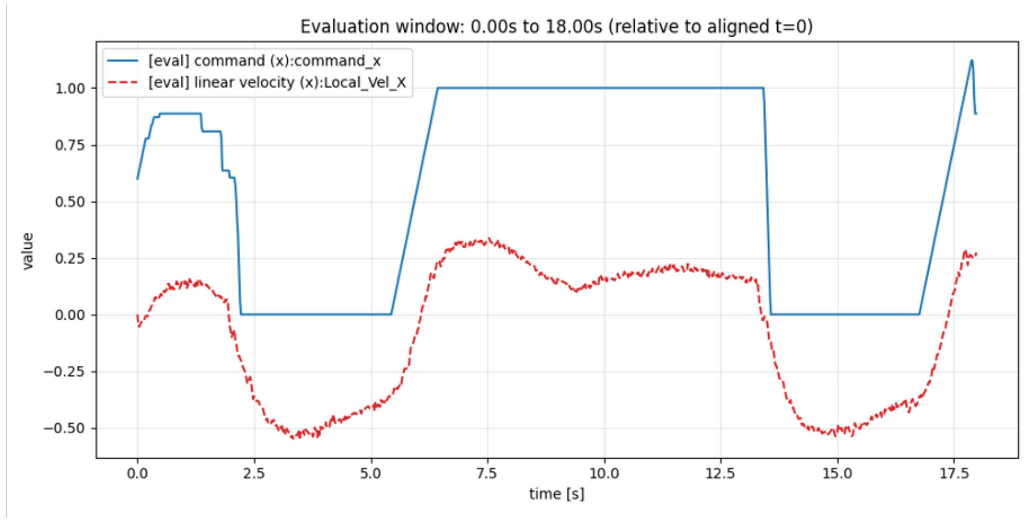
在评估时域内，SRM将均方根误差从0.673222 降低至**0.291831**，并将平均绝对误差从0.649399 降低至**0.230260**。此外，最后2秒内的稳态跟踪性能显著提升：w_{RMSE} 从0.631776 降低至**0.109146**，w_{MAE} 从0.596709 降低至**0.090787**。这些结果表明，SRM在硬件上同时提升了整体跟踪精度和最终调节行为。

定性行为

图4.4展示了SRM启用策略与基线的代表性真实世界试验，包括对齐的命令/响应轨迹及相关的误差汇总。尽管硬件运行之间的确切命令序列不同，但两次试验均展示了典型的阶梯式命令变化，并突出了跟踪行为的定性差异。在所示试验中，SRM启用策略表现出更小的跟踪偏差和更低的稳态偏差，这与表4.2中的定量比较结果一致。附录6.2中提供了更多实际世界图。

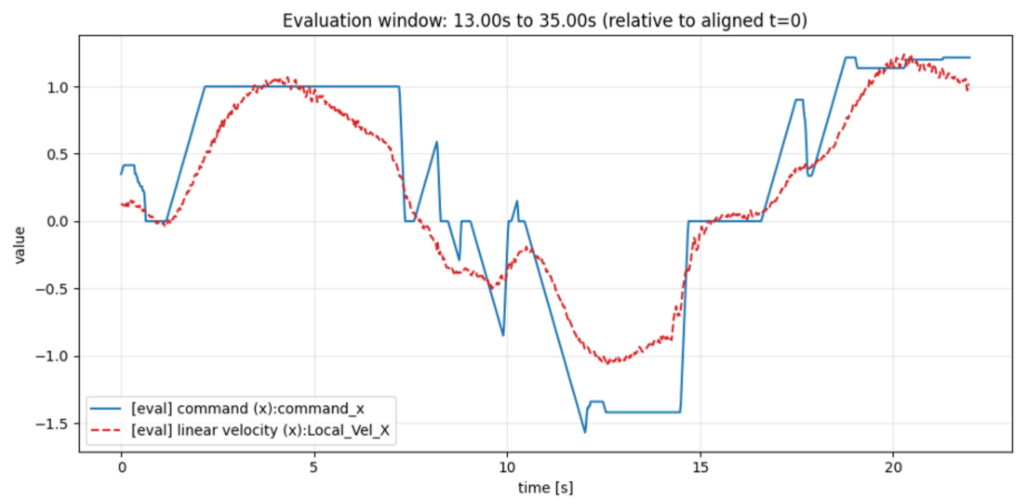


(a) with SRM (proposed).

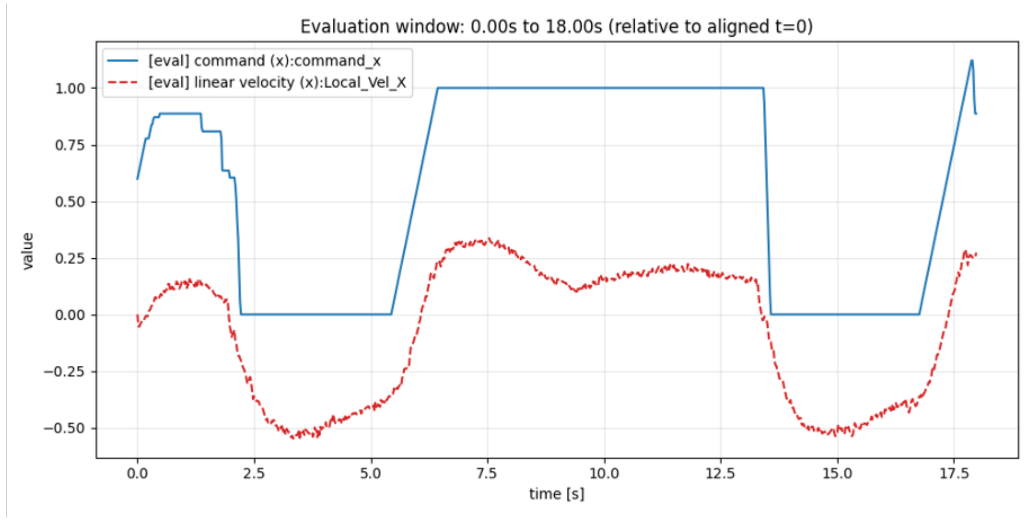


(b) without SRM (baseline).

Figure 4.4: Representative real-world forward-velocity tracking results. Full trajectories for additional trials are provided in Appendix 6.2.



(a) 使用SRM（本文提出）。

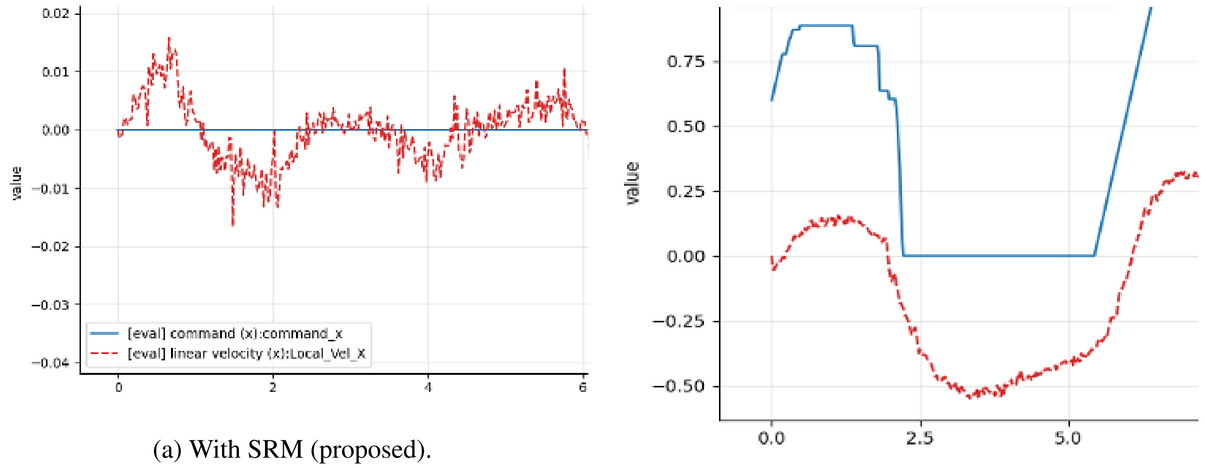


(b) 未使用SRM（基线）。

图4.4：具有代表性的真实世界前向速度跟踪结果。更多试验的完整轨迹见附录6.2。

Station-keeping at zero command

To further examine stability under zero command, we evaluate station-keeping behavior where the commanded forward velocity is held at $v_x^{\mathrm{cmd}} = 0$. Figure 4.5 shows representative real-world trials. With SRM, the measured forward velocity remains close to zero with small fluctuations, whereas the baseline without SRM exhibits noticeable drift, including sustained backward motion during the zero-command interval.



(a) With SRM (proposed).

(b) Without SRM (baseline).

Figure 4.5: Real-world station-keeping under zero command ($v_x^{\mathrm{cmd}} = 0$).

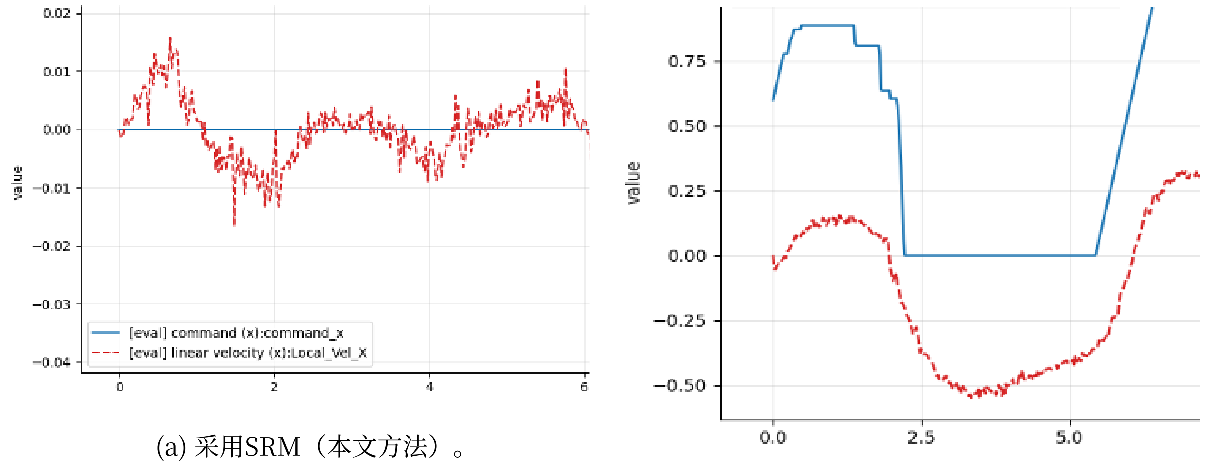
Table 4.3 summarizes tracking errors. The SRM-enabled policy achieves substantially smaller errors than the baseline, consistent with reduced drift in Fig. 4.5.

Table 4.3: Real-world station-keeping errors under zero command

Condition	**RMSE**	**MAE**
with SRM **(proposed)**	**0.013634**	**0.009517**
without SRM (baseline)	0.672472	0.694215

零指令下的站位保持

为进一步检验零指令下的稳定性，我们评估了指令前向速度保持在$v_x^{\mathrm{cmd}} = 0$时的定点保持行为。图4.5展示了具有代表性的真实世界试验。采用SRM时，测得的前向速度接近零且波动较小，而未采用SRM的基线则表现出明显的漂移，包括在零指令区间内持续的向后运动。



(a) 采用SRM（本文方法）。

(b) 未采用SRM（基线）。

图4.5：零指令下的真实世界定点保持（$v_x^{\mathrm{cmd}} = 0$）。

表4.3总结了跟踪误差。与基线相比，SRM启用策略实现了显著更小的误差，这与图4.5中漂移减少的结果一致。

表4.3：零指令下的真实世界定点保持误差

条件	**RMSE**	**MAE**
使用 SRM **（建议方案）**	**0.013634**	**0.009517**
不使用 SRM（基线）	0.672472	0.694215

V. Conclusion

In this study, we investigated sim-to-real deep reinforcement learning for hybrid wheeled–legged locomotion under partial observability. We introduced a *State Restoration Model (SRM)* that reconstructs a state-like latent representation from observation histories and used it to condition a Hybrid Locomotion Policy trained in simulation.

Across the proposed *simulation* → *simulation* → *real* evaluation pipeline, SRM improved commanded-velocity tracking and reduced steady-state bias compared with a baseline trained without SRM. In real-robot experiments, the SRM-enabled policy exhibited smaller tracking errors and more stable station-keeping behavior under near-zero commands, indicating that restoring a state-like representation from temporal context can mitigate ambiguity induced by partial observability and support more reliable transfer from simulation to hardware.

Several limitations suggest concrete directions for future work. The specific contribution of SRM-restored quantities beyond velocity-related effects has not been isolated, particularly because SRM approximates privileged full-state information that can include reward-related signals and contact- or height-related quantities; systematic ablations are needed to identify which restored components are responsible for the observed gains. In addition, the current real-world evaluation reflects nominal operating conditions, so broader stress tests (e.g., payload changes, surface/friction variations, and external perturbations) are required to characterize robustness envelopes and failure modes. Finally, the combination of reconstruction and reward-consistency objectives may introduce sensitivity to reward design and reward-scale mismatch between simulation and reality, motivating studies that reweight or remove the reward-consistency term and perturb reward components to assess behavioral stability.

V. 结论

在本研究中，我们探讨了部分可观测性下混合轮式-腿式运动的仿真到现实的深度强化学习。我们引入了一个状态恢复模型 *(SRM)*，该模型从观测历史中重建类状态潜在表示，并将其用于调节在仿真中训练的混合运动策略。

在提出的仿真 → 仿真 → 真实 评估流程中，与未使用SRM训练的基线相比，SRM改善了指令速度跟踪并减少了稳态偏差。在真实机器人实验中，SRM启用策略在接近零指令下表现出更小的跟踪误差和更稳定的驻留行为，这表明从时间上下文中恢复类状态表示可以缓解部分可观测性引起的模糊性，并支持更可靠的从仿真到硬件的迁移。

若干局限性为未来工作指明了具体方向。SRM恢复的量中，除速度相关效应外的具体贡献尚未被分离出来，这尤其因为SRM近似于特权全状态信息，其中可能包含奖励相关信号以及接触或高度相关量；需要进行系统性消融实验，以确定哪些恢复组件对观察到的增益负责。此外，当前的真实世界评估反映了标称运行条件，因此需要更广泛的压力测试（例如负载变化、表面/摩擦变化以及外部扰动）来表征鲁棒性包络和失效模式。最后，重建与奖励一致性目标的结合可能会引入对奖励设计以及仿真与现实之间奖励尺度不匹配的敏感性，这促使我们开展研究，通过重新加权或移除奖励一致性项并扰动奖励组件来评估行为稳定性。

VI. Appendix

6.1 Training Hyperparameters

Table 6.1 summarizes the main network architectures and PPO/SRM hyperparameters used in our experiments.

Table 6.1: Key training hyperparameters (network and optimization).

Category	**Setting**
Random seed	42
Rollout length	num_steps_per_env = 24
Training horizon	max_iterations = 5000
Policy/Value backbone	Actor–Critic MLP (ActorCritic)
Actor hidden dims	(512, 256, 128)
Critic hidden dims	(512, 256, 128)
Activation	ELU
Policy init noise	init_noise_std = 1.0
Algorithm	SRM-PPO (SRMPPO)
Learning rate	1×10^{-3} (adaptive schedule)
Discount / GAE	$\gamma = 0.99$, $\lambda = 0.95$
PPO clipping	clip_param = 0.2
Entropy coef.	0.01
Value loss coef.	1.0
Optimization	5 epochs, 4 mini-batches
KL target	desired_kl = 0.01
Grad clipping	max_grad_norm = 1.0
SRM backbone	GRU (srm_net=gru)
SRM input dims	srm_input_dim=32
SRM hidden dim	256
SRM output dim	5
SRM layers	1
History stacking	num_policy_stacks=2, num_critic_stacks=2

VI. 附录

6.1 训练超参数

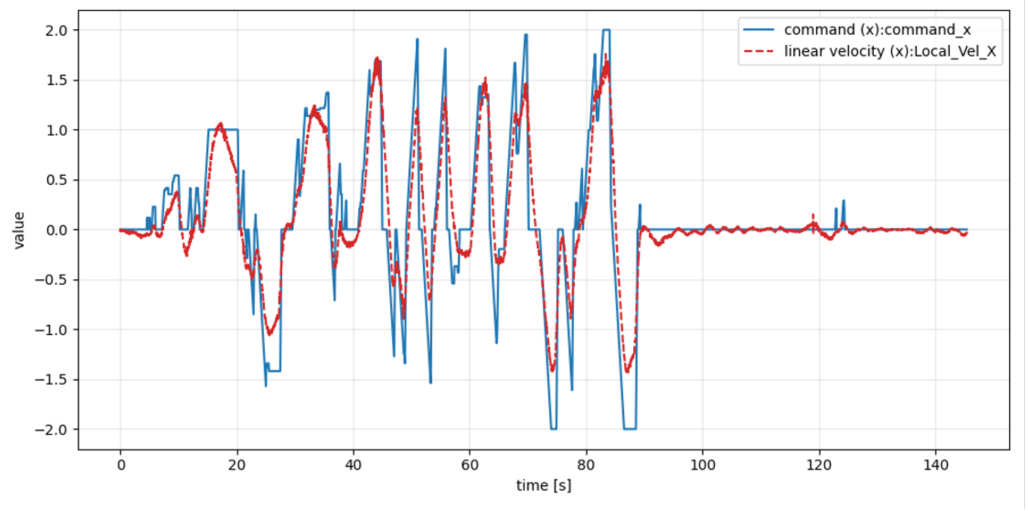
表6.1总结了实验中使用的网络架构以及PPO/SRM超参数。

表 6.1：关键训练超参数（网络与优化）。

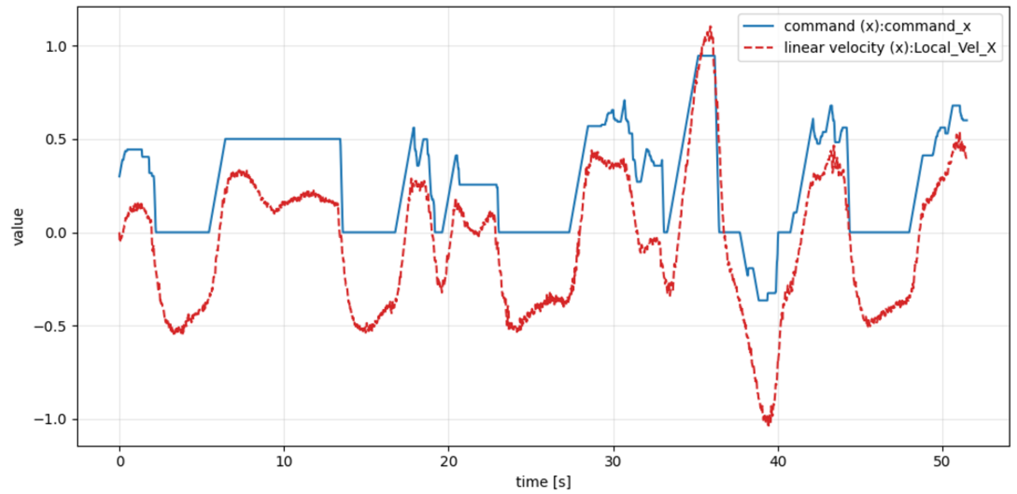
类别	**设置**
随机种子	42
展开长度	每个环境的步数 = 24
训练范围	最大迭代次数 = 5000
策略/价值主干网络	Actor-Critic MLP（演员-评论家）
Actor隐藏层维度	(512, 256, 128)
Critic隐藏层维度	(512, 256, 128)
激活函数	ELU
策略初始化噪声	初始化噪声标准差 = 1.0
算法	SRM-PPO (SRMPPO)
学习率	1×10^{-3} (自适应调度)
折扣因子/广义优势估计	$\gamma = 0.99$, $\lambda = 0.95$
PPO裁剪	裁剪参数 = 0.2
熵系数	0.01
价值损失系数	1.0
优化	5 轮次，4 小批量
KL目标	期望KL = 0.01
梯度裁剪	最大梯度范数 = 1.0
SRM主干网络	GRU (SRM网络=gru)
SRM输入维度	srm输入维度=32
SRM隐藏维度	256
SRM输出维度	5
SRM层数	1
历史堆叠	策略堆叠数=2，评论家堆叠数=2

6.2 Real-World Full Trajectories

This appendix provides additional full time-series trajectories for real-world experiments, supplementing the representative results shown in Section 4.2.



(a) With SRM (proposed).

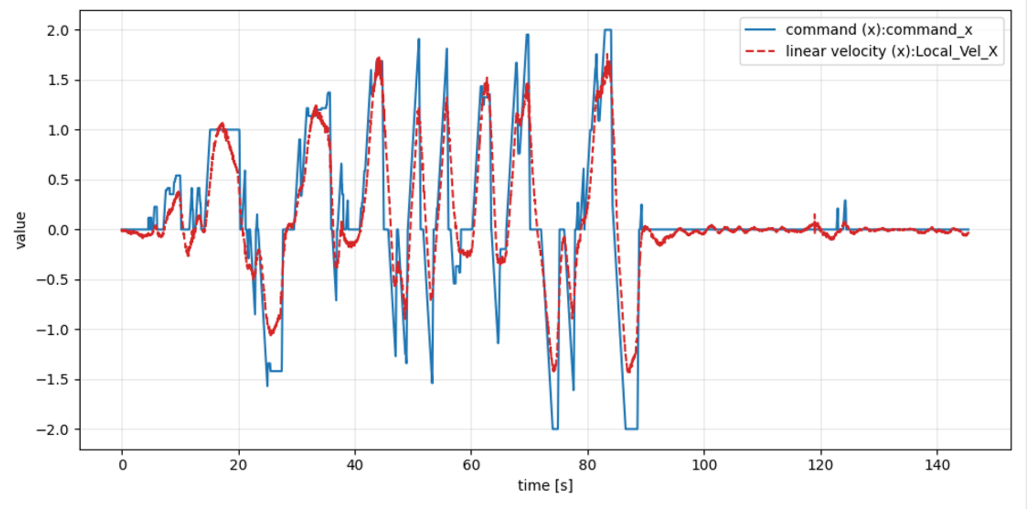


(b) Without SRM (baseline).

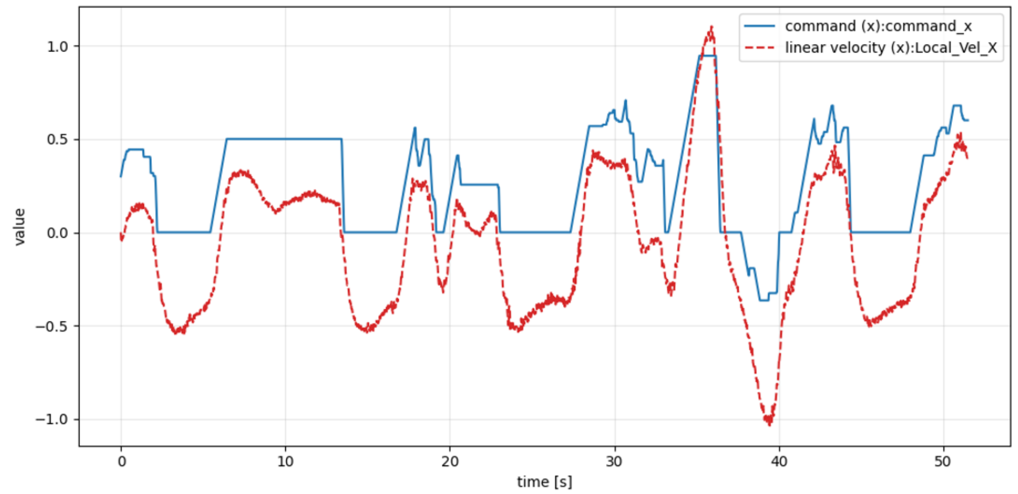
Figure 6.1: Full real-world trajectories used for qualitative reference.

6.2 真实世界完整轨迹

本附录提供了真实世界实验的额外完整时间序列轨迹，补充了第4.2节中展示的代表性结果。



(a) 使用SRM（本文提出的方法）。



(b) 未使用SRM（基线）。

图6.1：用于定性参考的完整真实世界轨迹。

6.3 Pseudocode for SRM-PPO Training

Algorithm 1 summarizes the training loop of the proposed SRM-PPO framework. At each iteration, we collect rollouts in simulation using the actor policy $\pi_\theta(a_t \mid s_t^o)$, where s_t^o denotes the partially observable input available at deployment. During training, the critic additionally receives privileged full-state information s_t^f that is accessible only in simulation. After collecting T steps, we compute returns R_t and advantages $\hat{A}_t$ using generalized advantage estimation (GAE) with (γ, λ).

During optimization, each mini-batch performs two coupled updates. First, the SRM parameters ψ are updated to reconstruct a privileged, full-state approximation $\hat{s}_t^f$ from an observation history $s_{t-k:t}^o$. The SRM objective combines a reconstruction loss that aligns $\hat{s}_t^f$ with s_t^f and a reward-consistency loss that aligns the reward computed from the reconstructed state with the reward computed from the privileged state:

$$\mathcal{L}_{\text{SRM}} = \mathcal{L}_{\text{reconstruction}}(\hat{s}_t^f, s_t^f) + \mathcal{L}_{\text{reward}}(R(\hat{s}_t^f, a_t), R(s_t^f, a_t)).$$

Second, the policy/value parameters θ are updated via PPO using the clipped surrogate objective, while the value function is trained to regress the return R_t using the critic input s_t^f during training. This alternating SRM update and PPO update encourages the actor to operate on a restored state representation while retaining the stability benefits of PPO.

6.3 SRM-PPO训练伪代码

算法1总结了所提出的SRM-PPO框架的训练循环。在每次迭代中，我们使用策略网络 $\pi_\theta(a_t \mid s_t^o)$在仿真中收集轨迹数据，其中$s_t^o$表示部署时可用的部分可观测输入。在训练期间，评论家网络额外接收仅在仿真中可访问的特权全状态信息 s_t^f。在收集 T 步后，我们使用广义优势估计（GAE）计算回报 R_t 和ˆ优势值 A_t，参数为(γ, λ)。

在优化过程中，每个小批量执行两次耦合更新。首先，更新SRM参数 ψ，以从观测历史 $s_{t-k:t}^o$中重构一个特权状态下的完整状态近似值ˆs_t^f。SRM目标结合了重构损失（使ˆs_t^f 与s_t^f 对齐）和奖励一致性损失（使从重构状态计算的奖励与从特权状态计算的奖励对齐）：

$$\mathcal{L}_{\text{SRM}} = \mathcal{L}_{\text{reconstruction}}(\hat{s}_t^f, s_t^f) + \mathcal{L}_{\text{reward}}(R(\hat{s}_t^f, a_t), R(s_t^f, a_t)).$$

其次，通过PPO使用裁剪替代目标更新策略/价值参数 θ，同时训练价值函数以回归回报 R_t，并在训练期间使用评论家输入 s_t^f。这种交替进行的SRM更新和PPO更新鼓励演员网络在恢复的状态表示上运行，同时保留PPO的稳定性优势。

Algorithm 1 SRM-PPO Training

Require: policy/value parameters θ, SRM parameters ψ, rollout horizon T, PPO hyperparameters $(\gamma, \lambda, \epsilon)$

```
while training do
    Initialize rollout buffer 𝒟 ← ∅
    for t = 1 to T do
        Observe actor input s_t^o (partial) and critic input s_t^f (privileged, sim only)
        Sample a_t ∼ π_θ(· | s_t^o) and execute in simulator
        Receive reward r_t and done flag d_t
        Store (s_t^o, s_t^f, a_t, r_t, d_t, log π_θ(a_t | s_t^o)) in 𝒟
    end for
    Compute returns R_t and advantages Â_t from 𝒟 using GAE(γ, λ)
    for each mini-batch ℬ ⊂ 𝒟 for several epochs do
      SRM update
        ŝ_t^f ← SRM_ψ(s_{t−k:t}^o)
        ℒ_reconstruction ← ℒ_Huber(ŝ_t^f, s_t^f)
        ℒ_reward ← ‖R(s_t^f, a_t) − R(ŝ_t^f, a_t)‖²
        ℒ_SRM ← ℒ_reconstruction + ℒ_reward
        Update ψ by minimizing ℒ_SRM
      PPO update
        ρ ← exp(log π_θ(a | s^o) − log π_old(a | s^o))
        ℒ_π ← 𝔼[max(−ρÂ, −clip(ρ, 1 − ϵ, 1 + ϵ)Â)]
        ℒ_v ← 𝔼[(V_θ(s^f) − R)²]
        Update θ by minimizing ℒ_π + ℒ_v
    end for
end while
```

算法1 SRM-PPO训练

需要：策略/价值参数 θ，SRM参数 ψ，轨迹收集步数 T，PPO超参数 $(\gamma,\ \lambda,\ \epsilon)$

```
当训练时执行
    初始化轨迹缓冲区 𝒟 ← ∅
    对于 t = 1到 T执行
        观察演员输入s_t^o（部分）和评论家输入 s_t^f（特权状态，仅限模拟器）
        采样 a_t ∼ π_θ(· | s_t^o) 并在模拟器中执行
        接收奖励 r_t 和完成标志 d_t
        将(s_t^o, s_t^f, a_t, r_t, d_t, 日志 π_θ(a_t | s_t^o)) 存储到 𝒟中
    结束循环
    计算回报 R_t 和优势值 Â_t 从 𝒟 使用GAE(γ, λ)
    对于每个小批量 ℬ ⊂ 𝒟 进行若干轮次 执行
  SRM更新
        ŝ_t^f ← SRMψ(s_{t−k:t}^o)
        ℒ_reconstruction ← ℒ_Huber(ˆs_t^f, s_t^f)
        ℒ_reward ← ‖R(s_t^f, a_t) − R(ˆs_t^f, a_t)‖²
        ℒ_SRM ← ℒ_reconstruction + ℒ_reward
        通过最小化 ℒ_SRM 来更新 ψ
  PPO更新
        ρ ← exp(log π_θ(a | s^o) − log π_old(a | s^o))
        ℒ_π ← 𝔼[max(−ρˆA, −clip(ρ, 1 − ϵ, 1 + ϵ)ˆA)]
        ℒ_v ← 𝔼[(V_θ(s^f) − R)²]
        通过最小化 ℒ_π + ℒ_v 来更新 θ
    结束循环
结束当型循环
```

요 약 문

실세계 로봇에서 보행 정책을 안정적으로 실행하는 것은 제한된 센서 관측으로 인해 상태를 완전하게 관측하기 어렵고, 시뮬레이션에서 학습한 정책이 시뮬레이션–현실 격차로 인해 실제 하드웨어에서 성능이 저하될 수 있다는 점에서 여전히 도전적이다. 본 논문은 바퀴와 다리를 결합한 하이브리드 주행을 수행하는 이륜-이족 로봇을 대상으로, 시뮬레이션에서 학습한 심층 강화학습 정책을 실제 로봇에 전이하기 위한 방법을 다룬다. 이 문제는 바퀴–다리 결합 동역학과 간헐적 접촉으로 인해 부분가시성 특성이 두드러지며, 단일 시점의 관측만으로는 안정적인 제어에 필요한 마르코프 상태를 구성하기 어렵다.

이를 해결하기 위해, 관측 이력으로부터 상태에 준하는 잠재표현을 복원하는 상태 복원 모델(State Restoration Model, SRM)을 제안한다. SRM은 시뮬레이션에서만 이용 가능한 특권 정보(전체 상태)를 근사하도록 학습되며, 비대칭 액터–크리틱(asymmetric actor–critic) 학습 구조에 통합된다. 학습 시 크리틱은 특권 정보를 활용해 정책을 평가하고, 액터는 SRM이 복원한 표현을 입력으로 받아 실제 환경에서도 일관된 의사결정을 수행하도록 한다.

제안 방법은 FlaminGO 8자유도 이륜-이족 로봇에서 검증하였다. Isaac Sim 에서 정책을 학습한 뒤, 독립 물리 엔진인 MuJoCo에서 교차 시뮬레이터 검증을 수행하고, 최종적으로 실제 로봇에 배포하는 시뮬레이션 → 시뮬레이션 → 현실 절차로 평가하였다. 실험 결과 SRM을 적용한 정책은 SRM이 없는 기준 방법 대비 속도 명령 추종 성능이 향상되고 정상상태 편향이 감소하였으며, 특히 영(0)에 가까운 명령에서의 정지 유지 동작이 더 안정적으로 나타났다. 이는 시간적 문맥을 활용해 상태에 준하는 표현을 복원하는 접근이 부분가시성으로 인한 불확실성을 완화하고, 하이브리드 주행에서의 시뮬레이션–현실 전이 강건성을 향상시킬 수 있음을 시사한다.

요 약 문

在真实世界机器人中稳定执行行走策略仍具挑战性，原因在于受限的传感器观测导致难以完全观测状态，且仿真中学习的策略可能因仿真-现实差距而在实际硬件上性能下降。本论文针对执行轮腿混合驱动的两轮双足机器人，探讨如何将在仿真中学习的深度强化学习策略迁移至真实机器人。该问题因轮-腿耦合动力学与间歇性接触而凸显部分可观测性特征，仅凭单次观测难以构建稳定控制所需的马尔可夫状态。

为了解决这一问题，本文提出了一种状态恢复模型（State Restoration Model, SRM），该模型能够从观测历史中恢复出与状态相当的潜在表示。SRM 通过学习近似仅在仿真中可用的特权信息（完整状态），并集成到非对称演员-评论家（asymmetric actor–critic）学习框架中。在训练过程中，评论家网络利用特权信息评估策略，而演员网络则接收 SRM 恢复的表示作为输入，从而在真实环境中也能做出一致的决策。

所提出的方法在FlaminGO 8自由度两轮双足机器人上进行了验证。在Isaac Sim中学习策略后，在独立物理引擎MuJoCo中进行交叉仿真器验证，并最终通过从仿真 → 到仿真 → 再到现实的流程部署到实际机器人上进行评估。实验结果表明，应用SRM的策略相比未使用SRM的基准方法，在速度指令跟踪性能上有所提升，稳态偏差减小，尤其是在接近零指令时的静止保持行为更加稳定。这表明，利用时间上下文恢复近似状态表征的方法能够缓解部分可观测性带来的不确定性，并提升混合驱动模式下仿真-现实迁移鲁棒性。

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먼저, 연구에 대한 깊은 통찰력과 따뜻한 인품으로 부족한 제게 학문의 길을 열어주신 한수희 교수님께 깊은 존경과 감사의 말씀을 올립니다. 교수님께서 보여주신 연구에 대한 열정과 세심한 지도편달은 제가 연구자로서 갖춰야 할 태도를 배우는 큰 밑거름이 되었습니다. 앞으로 나아갈 길에서도 교수님의 가르침을 잊지 않고 정진하는 연구자가 되겠습니다. 또한, 바쁘신 일정 중에도 흔쾌히 학위 심사를 맡아주시고, 제 연구가 한 단계 발전할 수 있도록 아낌없는 조언을 해주신 김상우 교수님과 박부견 교수님께도 진심으로 감사드립니다.

동고동락하며 연구실 생활의 큰 힘이 되어준 CoCEL 식구들에게도 고마운 마음을 전합니다. 연구 내외적으로 든든한 버팀목이자 코셀로의 CTO 인공지능 대가 장원 형, 코셀로 이사이자 저의 소중한 포항 메이트 상현이, AI팀 팀장이자 든든한 버팀목인 종수 형, 너무나 착한 지석이 형, 함께 밤을 지새우며 도와주었던 민재와 준영이, 수업의 든든한 버팀목이 되어준 제어 전문가 동형이, 로봇으로 마음이 통하는 수학자 민종이, 마지막 실험까지 발 벗고 도와준 매니폴드 장인 진우, 함께한 추억이 많은 승현 형, 드론팀 팀장이자 든든한 인간 GPT 지호, 그리고 연구실에 새로운 활력을 불어넣어 주는 성민, 준호, 근영, 혜령, 든든한 신일 형님 등 모든 동료 덕분에 힘들었던 순간들도 웃으며 이겨낼 수 있었습니다. 서로 다른 연구 분야임에도 세미나를 통해 귀한 아이디어를 나누고 격려해 준 드론팀, SLAM팀, 에너지팀 분들에게도 감사를 표합니다.

마지막으로, 저를 향한 무한한 믿음과 사랑으로 언제나 제 편이 되어주시는 아버지, 어머니께 말로는 다 할 수 없는 감사의 마음을 바칩니다. 그리고 항상 묵묵히 저를 응원해 주는 우리 형에게도 고마움을 전합니다. 우리 가족의 앞날에 건강과 행복이 가득하기를 기원합니다. 가족들의 헌신적인 지지가 있었기에 흔들리지 않고 학업에 매진할 수 있었습니다. 언제나 든든한 아들이자 동생이 되어 그 사랑에 보답하겠습니다.

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